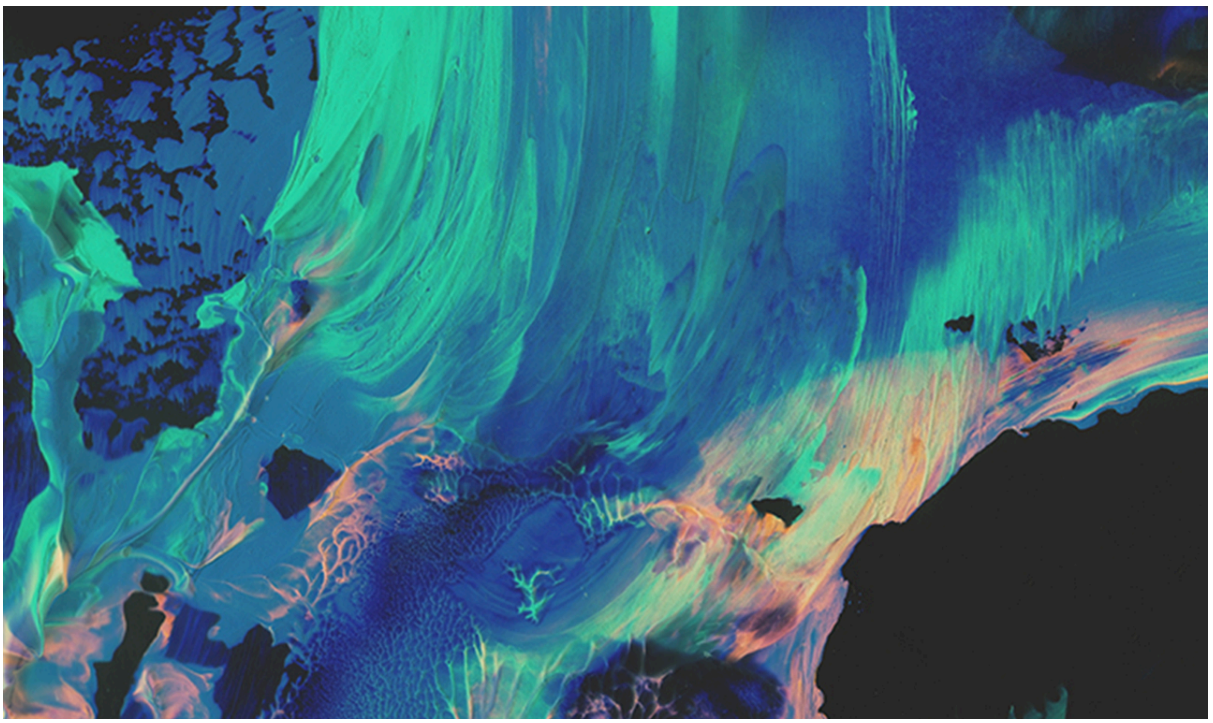


Are low emission zones effective?

An event study examining Swedish municipalities

Jon Rikner

Master Thesis in Cultural Geography - Umeå University

Spring 2026

Acknowledgement

I would like to express my gratitude to my supervisor, Marcin Rataj, for their guidance, constructive feedback, and continuous support throughout this thesis. Their expertise and encouragement were very helpful and insightful, which influenced the shaping of this research.

I would also like to thank the faculty and staff at Umeå University for providing a stimulating academic environment and the resources necessary to complete this work.

Special thanks go to my fellow students and friends for their support, discussions, and encouragement during the writing process.

Finally, I would like to give my biggest thanks to my family for their unwavering support, patience, and belief in me throughout my studies. Their encouragement has been a constant source of motivation.

Abstract

Air pollution from road traffic remains a major public health and environmental challenge in urban areas, and low emission zones (LEZs) have become widely adopted regulatory responses. In Sweden, where municipalities have substantial autonomy over their implementation, the effectiveness of these zones has received limited systematic empirical attention. This thesis examines whether LEZs are effective by analysing three Swedish cities Umeå, Helsingborg, and Uppsala that introduced such zones recently enough to enable credible pre- and post-implementation comparisons. Using annual data from Statistics Sweden (SCB) and the Swedish Meteorological and Hydrological Institute (SMHI), the study combines descriptive spatial analysis in ArcGIS with a staggered event study difference-in-differences design. It estimates effects on nitrogen dioxide (NO2) concentrations, privately owned cars per 1,000 inhabitants, and heavy lorry traffic relative to a comparable control group. The findings are interpreted through governance, policy implementation, and urban politics frameworks. The results show a clear and statistically significant reduction in privately owned cars per capita following implementation, suggesting a durable behavioural effect. NO2 concentrations exhibit weak and less robust indications of decline over time, while effects on heavy lorry traffic are statistically inconclusive, however, this may be due to the level of policy implementation. Descriptive trends also reveal declining diesel vehicle shares and rapidly increasing electric vehicle adoption in the studied cities, although the causal role of LEZs in this transition remains unclear. Overall, the evidence suggests that LEZs are most effective in shaping individual car ownership behaviour and represent a promising but partial instrument for improving urban air quality.

Keywords: Low emission zones, environmental policy effectiveness, event study, difference-in-differences, urban air quality, Swedish municipalities, NO2, car ownership

1. Introduction **1**
1.1 Aim 5
2. Background **6**
3. Theoretical framework **9**
3.1 Governance theory 10
3.1.1 Environmental Policy Effectiveness 10
3.1.2 Effectiveness 12
3.1.3 Environmental impact indicators 13
3.2 Policy implementation theory 15
3.2.1 Top down approach 16
3.2.2 Bottom up approach 17
3.3 Urban politics 17
3.3.1 Command and control 18
3.3.2 Market oriented strategy 19
4. Data and Method **19**
4.1 Data 20
4.1.1 Variables 20
4.1.2 Nitrogen dioxide 21
4.1.3 Total number of cars and cars per capita 21
4.1.4 Heavy lorries 22
4.2 Methodology 22
4.2.1 Descriptive analysis 22
4.2.2 Event study 24
4.3 Model specification 27
4.4 Ethical considerations 29
4.5 Limitations 30
5. Results **31**
5.1 Descriptive findings 32
5.2 Event study analysis 40
6. Discussion **47**
6.1 Are the low emission zone policy showing effectiveness? 47
6.2 Does the policy implementation work well? 51
6.3 Urban politics on low emission zone 52
6.4 Future Research 55
7. Conclusion **55**
7.1 The implementation of low emission zones in Swedish cities influence on changes in car ownership 56
7.2 Low emission zones influence on the composition of vehicle fleets 57
7.3 Low emission zones affect on environmental outcomes, behavioral changes, and technological adaptation 58
8. References **61**
9. Appendix **74**

1. Introduction

Air pollution is a growing concern worldwide, with a mortality rate at 7 million people annually demonstrates the magnitude of this challenge (UN, 2023). There are numerous health risks generated by air pollution, with vehicles and especially road vehicles being a central element of emitting adverse pollution (Long & Carlsten, 2022). Naturvårdsverket (2025) further elaborates on the opportunity to recover or develop a sustainable solution, to reduce the greenhouse emissions to a required level, changes within the transport sector from fossil fuels is prerequisite. Thus, having varying opportunities depending on geographical location, an intervention is of highest importance (Naturvårdsverket, 2025). One solution is to use low emission zones in cities to reduce vehicles with high emission to increase the air quality. The growing need for targeted intervention naturally leads to a closer assessment of low emission zones as a policy tool.

Building the need for effective interventions, low emission zones is discussed as one of the most prominent and widely discussed regulatory approaches. Low emission zones, a restriction in cities used to terminate the usage of vehicles of certain emission classes (Broster & Terzano, 2025; Transportstyrelsen, 2024). In Sweden, the municipalities are responsible for implementing zones that have

requirements on vehicles and decide upon limits for where they are allowed to drive (Transportstyrelsen, 2024). There are different types of environmental zones, having different requirements on what vehicles and how long they are allowed to drive where (Umeå kommun, 2020). This has led to companies doing testing fraudulently to manipulate the classing of certain (Broster &Terzano, 2025). Broster & Terzano (2025) further argue that political debates whether implementation of low emission zones is required is common, this in turn can lead to different municipalities introducing these zones, depending on what parties have local power. The political dynamics emphasizes the importance of long term planning, and stable governance structures.

Given the political variation surrounding environmental zones, long term national goals become even more foundational. Naturvårdsverket (2025) is providing an idea to reduce domestic transportation emissions by 70%. However, this will take a long time and extensive work (Naturvårdsverket, 2025). To be able to create a long-term plan the politicians need to have a cross-party goal or a party that is able to be a majority for several election periods (Leruth, 2024). Leruth (2024) explains the need of focusing on long term goals rather than short term outcomes to more effectively overcome urgent and great crises. This need for political stability also forms how environmental policies are implemented at the local level.

At the municipal level, political leadership becomes a critical factor in determining environmental outcomes. Local politics have a central role in low emission zone regulation, thus the party with majority in Swedish municipalities have a significant influence. The parties decide on green politics in respective municipalities. At the same time as the local success is important, surrounding cities accomplishments in their policies (Lindvall, 2023). Van der Heijden (2019) elaborates on local politics, and clarifies that cities are both playing the key role in adjustments and mitigation. These local dynamics further intersect with broader urban development patterns and transportation tendencies.

As cities continue to evolve and grow, transportation patterns increasingly shape both environmental and public health outcomes. Road transport is enhancing air pollution and is a critical polluter in urban areas. The transport is creating critical air pollution and environmental pressure, in turn the public health is of concern (Tafidis et al., 2024). Thana Singam et al., (2024) elaborates on the importance of well planned cities and land use to create sustainable cities, the reduction of light-duty private vehicles fosters sustainable cities. The importance of lowering the number of light-duty vehicles constructs the need for low emission zones (Thana Singam et al., 2024). Transport vehicles have improved possibilities to move further from the city without removing all the

advantages of living close to an urban center, this in turn increases the use of vehicles in urban areas (Ostermeijer et al., 2022). Ostermeijer et al., (2022) continues to explain that one more car per 100 inhabitants reduces population density by 2,2%. However, the movement is often to build up areas surrounding the city and rather than increasing ruralisation, it makes cities larger and the need for owning vehicles greater (Wu et al., 2023). These tendencies underscores why understanding vehicle composition in different cities is fundamental for designing effective environmental policies.

Considering the close link between vehicle use and environmental impact, examining vehicle characteristics becomes a crucial next step. The number of vehicles are of high relevance for emissions and environmental cities. Thus, it is important to note that different types of vehicles have different impacts and emissions. Therefore, it is interesting to investigate what type of vehicles are most prominent in each city, both by type of vehicle and type of fuel. This examination provides the foundation for evaluating how low emission zones and related policies can be refined to be tailored for local conditions.

1.1 Aim

The aim of this study is to analyze the low emission zones and their impact in Sweden in three different cities; Umeå, Helsingborg and Uppsala. Given the high autonomy of Swedish municipalities the understanding of varying local politics as this may affect effectiveness of the policy. This in turn is also central for explaining variation in climate policy at a local level and implementation of local sustainable transport measures. Therefore, environmental, behavioral and technological outcomes are very relevant in low emission zones in Sweden and potentially affect both the total number of cars or fuel type on a local but also a more general and nationwide level. These outcomes in turn influence the emission levels and the pollutants in the air, and to improve air quality is the main objective of low emission zones. To answer the aim, the following three research questions has been set:

- To what extent are low emission zones in Swedish cities associated with changes in the total number of cars and cars per capita over time?

- How do low emission zones influence the composition of vehicle fleets, particularly the share of different fuel types (e.g., fossil vs. alternative fuels)?

- To what extent do low emission zones affect environmental outcomes, behavioral changes, and technological adaptation?

2. Background

Low emission zones generally lowers NO2, and with that improve air quality, public health and quality of life (Bruyneel et al., 2025). Previous research emphasises on how low emission zones improve air quality and quality of life. Low emission zones focus on several aspects of reducing emissions and could be interpreted differently, but there is one main goal: to improve air quality and public health. The low emission zones promote a cleaner transport sector, this includes a mobility shift from cars towards walking and biking. At the same time, a shift in transport fuel is important, as some travel requires motorised vehicles. Low emission zones also have economic and social aspects, as financial factors need to generate sufficient benefits to make investments in LEZs worthwhile without compromising inhabitants' well being. Sarmiento et al., (2023) further elaborates on the importance of clean air for life satisfaction, but in some instances is the access to transport outperforming air quality for life quality. Meaning that the behavioural aspect is important, as inhabitants' willingness to adapt, and the extent to which the policy interferes with everyday life, affect quality of life.

To improve life quality there is a need for a combination of working outcomes of the implemented policy intertwined with behavioral outcomes that aligns with the aim of the low emission zone policy. Due to the fact that the world is

trending towards overconsumption and less environmentally friendly, policies like low emission zones could potentially be a way of fighting this problem. To determine whether low emission zones counteract negative environmental trends, it is necessary to test whether they are effective.

Low emission zones are not a policy tool uniquely used in Sweden, it is rather a part of a wider global trend aimed at improving air quality and reducing air pollution in urban areas connected to road transportation. Many European cities have implemented low emission zones over the last few decades (Tögel and Špička, 2014). Similar policy tools have also been implemented in parts of Asia, especially in Chinese urban areas. This for improving both local and global air quality and improving public health.

While the overall goal is clear and simple; to improve public health and air quality by reducing emissions, the design and effectiveness can vary depending on governance structure and local geographies. Some cities rely on strict vehicle access restrictions, often based on emission standards, while others have broader or slimmer limitations to mobility and transport. The result of this creates heterogeneous effects depending on behavioral change, economic impacts and observed outcomes. This means the policy is context based and cannot be directly copied from one country nor a city in the same country exactly into another. The policy cannot be generalised without taking behavioural and structural differences into account.

Sweden has eight municipalities that have implemented low emission zones (Trafikverket, 2024) and four of them were already introduced in the 20th

century, meaning that they have been implemented for a long time. Therefore, it is difficult to get data prior to the implementation, or enough data to create robust results. Although they have been implemented for a long time the urgent environmental problems have grown stronger in the most recent decades. Along with three of the four early implementations being Sweden's three largest cities: Stockholm, Gothenburg, and Malmö, these are often difficult to compare with other cities due to their size. It is therefore more informative to focus on and compare cities that are more similar in order to minimise the influence of external factors on observed changes rather than attributing them to low emission zones. Larger cities are generally more affected by other variables to a greater extent. Having the more recent implementation with more robust that along with a higher relevance to today's politics funneled the research towards cities with more recent implementation. This leads to the importance of looking at the municipalities that have implemented low emission zones to tackle air quality and provide better public health.

Previous research on low emission zones in Sweden has primarily focused on their associated health benefits rather than their effectiveness as a policy instrument for reducing emissions. In particular, existing studies have examined the direct health outcomes of improved air quality following implementation. For example, the study by Flanagan et al. (2022) focuses primarily on Malmö, which limits the understanding of whether the observed effects are unique to that city or whether similar outcomes would be observed in other urban contexts. Consequently, the extent to which the effectiveness of low emission zones varies across different Swedish cities remains insufficiently explored.

Although there is research on effectiveness across European cities including cities in Sweden (generally the three biggest ones; Stockholm, Gothenburg and Malmö if Sweden are included) and other aspects that low emission zones may affect in Sweden. There is still no broader research on low emission zones and its effectiveness in Sweden, especially focusing on the smaller cities and municipalities that have implemented the policy. It is therefore interesting to test to what extent low emission zones impact different Swedish municipalities on environmental, behavioral and technological adaptation.

In contrast to previous research in Sweden that has primarily focused on individual cities, this study adopts a broader geographical scope by including multiple urban areas. This approach facilitates a more comprehensive assessment of the effectiveness of low emission zones and enhances the generalizability of the findings. For this reason, the present study contributes to the existing body of knowledge on the subject by providing additional evidence regarding the effectiveness of low emission zones in Sweden. In doing so, it offers valuable insights into the extent to which these policies achieve their intended environmental objectives across different urban contexts.

3. Theoretical framework

This chapter focuses on prior research and theories, there will also be important concepts introduced in this section.

3.1 Governance theory

Governance can take multiple roles and forms, under conditions of high municipal autonomy, it plays a key role in the implementation of local policies, and the relationship between governance style and policy can influence policy outcomes (Filgueiras et al., 2023). To achieve success when implementing a policy, the implementation should be carefully implemented with focus on the current geographical location and its attributes to provide the best and smooth coexistence with other policies along with the cities needs (Vince et al., 2024). This is set to nurture that the practical tools are used to achieve goals effectively, such as subsides, taxes and/or regulations. The main goal is to provide a functioning implementation to create a successful policy and further make the policy effective. For the governance to be successful the carried out implementations needs to work, and be controlled for. This highlights the importance of evaluating whether or not governance and implementation strategies actually produce the intended environmental outcomes.

3.1.1 Environmental Policy Effectiveness

To assess these outcomes, the concept of environmental policy effectiveness becomes central in understanding how well implemented policies achieve their objectives. Environmental policy effectiveness refers to the extent to which policies achieve their intended objectives. In this research, it is used to evaluate the low emission zone policy, and environmental policies are considered instruments for achieving sustainable development (Ioannou et al., 2025). The low emission zone policy is implemented to reduce emissions and improve air quality, while also serving as a strategy for sustainable development. The environmental policy effectiveness theory does not specifically focus on the policy itself, but rather evaluates how its implementation functions. This is

important for bridging the gap between policy design and performance (Skjærseth & Wettestad, 2010). The principle of environmental policy effectiveness is appealing on the conceptual level, the transition towards action does not always reflect the policy and some trade offs will always be inevitable (Nilsson & Persson, 2003; Naturvårdsverket, 1999). Low emission zones (LEZ) are a regulatory instrument aimed at reducing the use of heavy-duty vehicles in order to decrease road transport emissions and improve air quality. The policy contributes to sustainable development by restricting or incentivising vehicles that do not meet environmental standards (Transportstyrelsen, 2024).

If the policy is effective, its performance should accordingly achieve the stated goals. However, it is important to note that effectiveness is rarely absolute; achievements are always relative to place and context (Handoyo, 2024). Handoyo (2024) also highlights the importance of ensuring that data is presented objectively and not overly positively framed by governments, in order to avoid overstating policy effectiveness and to ensure a fair assessment.

To measure the effectiveness of low emission zones, emissions generated by vehicles need to be assessed, as well as the types of vehicles used and the number of cars before and after the policy was implemented. In this research environmental policy effectiveness will be measured by NO2 emission, number of cars, number of heavy lorries and fuel type. Seelarbokus (2014) elaborates on limitations in both data and understanding of trade offs in effectiveness, implementation, and geographical circumstances are highly relevant and need to be taken into consideration. It is therefore valuable to include more than one geographical location in order to provide a more generalisable understanding when measuring policy effectiveness.

3.1.2 Effectiveness

Effectiveness focuses on the outcomes of a policy, in contrast to efficiency, which focuses on achieving results with as little effort as possible (Otola, 2010). In this research, effectiveness focuses on whether LEZs reduce emissions, improve air quality, decrease the number of cars per capita, and shift fuel types towards more environmentally friendly alternatives. Effectiveness is important to measure, because if implementation is not effective and does not produce the desired outcomes, it may be unnecessary to implement, and the associated costs can be avoided (Eisman et al., 2020).

However, it is important to be aware of other factors that may influence the outcomes. Effectiveness can be assessed across multiple dimensions: improved access to certain types of cars can influence fuel type, population size will affect the number of cars, and other policies as well as access to public transport may also impact total emissions. In this study, effectiveness will be operationalised using the indicators are divided into three categories: environmental outcomes (emissions, air quality), behavioural outcomes (car ownership/use), and technological outcomes (fuel type). These are different aspects that need to be taken into consideration, as they influence effectiveness.

Therefore, while effectiveness provides a useful framework for evaluating environmental policy performance, it is also important to understand it as context dependent and relative to the specific situation, rather than as an absolute measure of favourable outcomes (Mickwitz, 2003). Thus, the achievement of policy goals can still indicate a meaningful impact, particularly

in complex urban areas where multiple transport systems and policies interact, Eisman et al., (2020) emphasises the fact that requiring quality evidence based considerations are needed before copying a policy into another place.

Effectiveness cannot be viewed in strictly binary terms of good or bad, nor can a policy. It is important to understand the policy in order to assess what is working and what can be improved (Smith & Stirling, 2007). Partial effectiveness may be observed, whereby emissions are reduced and the number of vehicles not meeting the criteria decreases, without the policy fully achieving its overall goals (Puig et al., 2025). By partly reaching its goals, an assessment can be made of future plans and how to improve the effectiveness of the policy (Shin & Lee, 2025; Manski, 2011). Partial effectiveness is often influenced by regulatory design; overly or under-regulated policies may fail to achieve their intended goals, therefore measures of effectiveness are needed.

3.1.3 Environmental impact indicators

Environmental impact indicators are measurable factors used to assess changes in environmental conditions and impacts, in order to evaluate the function and outcomes of a policy intervention (European Environment Agency, n.d; OECD, 2025).. Impact indicators provide a means of observing and quantifying conceptual policy goals, such as NO2 levels, air quality and economic performance, in order to assess outcomes (OECD, 2025; Gudmundsson, 2003). The environmental impact indicators further assess the ecological impacts of human activities.

When analysing policies, indicators play a central role in evaluation and operationalisation by connecting policy actions and implementation to measurable outcomes (Gudmundsson, 2003; European Environment Agency, n.d). The indicators are designed to provide insight into how natural habitats are affected by the environmental impacts of policies and other interventions. They enable the tracking of changes over time, comparison across policies and locations, identification of critical areas, and support improved decision-making and policy development (Bateman et al., 2013).

In the context of LEZs, requirements are imposed on drivers, which are intended to reduce emissions, as vehicle emissions place pressure on the environment and subsequently affect air quality. For this reason, LEZs are implemented as a policy response to this issue. In this research, environmental impact indicators are categorized into three groups, environmental indicators (emissions, air quality), behavioural indicators (car ownership/use), and technological indicators (fuel type), which all capture head on environmental impacts of LEZs policies. The environmental outcomes include both short and long term effects, whereby the policy has a direct impact that may evolve over time. The behavioural indicators reflect mobility patterns influenced by the implemented policy. The technological indicators, represented by fuel type, capture long term structural changes resulting from the LEZ policy, as LEZs aim to promote a shift towards more environmentally friendly and low emission vehicles.

However, there are limitations. The indicators that quantify abstract policy goals can post as oversimplifying complex systems and environmental systems can also be affected by other conditions (Mickwitz, 2003; Gudmundsson, 2003).

Meaning that the contributor to change is not always the policy itself but rather in combination with factors like weather, economic circumstances and trends as well as other policies. Therefore, it could be difficult to identify and evaluate the LEZs policy itself, and often the result can be overly positive and the effectiveness is higher measured than it actually is. Due to certain outcomes being linked to the LEZs policy and other factors are not fully taken into consideration.

This makes environmental impact indicators a great tool to use and is a great way to quantify, and an essential tool to evaluate policies. The results of the outcomes however need to be carefully evaluated and knowledge of other factors needs to be taken into consideration for potential influencing impact as well as other limitations in data that can arise.

3.2 Policy implementation theory

Policy implementation analyses how government policies are translated into action, focusing on how effectively they are carried out, rather than solely on their outcomes (Mathison, 2005; Grindle, 1980). The risk of an implementation gap in policy implementation is highly important to understand, as if the intended policy goals are not matched by actions to achieve those outcomes, the policy will fall short. For policies to be implemented effectively, effective communication is required. Communication between policymakers and implementers is crucial to overcoming the challenges of translating theory into practice, ensuring that implementation proceeds as smoothly as possible and supports the achievement of the intended policy objectives (Bullock et al., 2021). Purtle et al., (2023) elaborates on how policies also play a dual role: on

the one hand, they are implemented to achieve specific goals (e.g., reducing emissions), and on the other hand, they provide general knowledge about the implemented policy and support evidence based functions in implementation. Policy implementation is often divided into two main approaches: top down and bottom up.

3.2.1 Top down approach

The top down approach centres on goals and policies decided at higher levels (i.e., EU or government level), which are then cascaded through hierarchical structures for implementation. The top down approach emphasizes the "big picture" at a macro level, with standardized procedures focusing on overarching goals and the development and implementation of policies designed to achieve those objectives (Li, 2023). Li (2023) continues, high level leadership is oriented towards applying uniform strategies and operating on a broader scale, prioritising what is effective at a larger level. This approach may not fully account for local conditions. However, its rationale lies in providing clear direction and promoting administrative efficiency.

The limitations of the top down approach lie in its limited integration with those directly affected by the issues, as well as its tendency to generalise problems that may require different solutions depending on local needs (Heyden et al., 2016). Therefore, it is important to acknowledge the difference between functioning implementation and effectiveness of implementation.

3.2.2 Bottom up approach

The bottom up approach focuses on developing ideas, strategies, decision-making, and planning from lower levels, rather than from top management. The bottom-up approach promotes and values innovation, as well as detailed ideas and strategies from those directly involved in the issue or task, while also engaging all levels of employees or citizens to create a feeling of ownership and proactive identification of risks (Kaiser, 2020). Kaiser (2020) continues, the bottom up approach focuses on specific, localised issues, developing solutions at the margins that gradually work inward to address the core problem.

The limitations of the bottom-up approach include the potential for slower decision-making, as involving many stakeholders can delay action. Sabatier (1986) explains another risk of decision making driven by those directly involved is that it may overlook the broader issue, failing to engage with the "big picture" and potentially not fully resolving the problem. In addition, the bottom up approach is heavily reliant on the participation and perceptions of its participants.

3.3 Urban politics

Urban politics theory studies how collaboration and power relations shape urban areas. Power and collaboration in a city influence social equality and inequality, economic development, and urban growth, with governance playing a central role in determining the direction of these developments. Politics is an ongoing activity present in almost all contexts, and all individuals are, in some form, part of this process, which is reproduced through everyday social practices and ways

of living (Stoker, 1998). The city functions as a coalition, driven by its population and shaped through both formal and informal alliances between public officials and private interests (Sutherland, 2025). Within urban politics, the political leadership is the central determinant when it comes to municipal policy implementation, and environmental regulations (Lindvall, 2023). Political parties pursue different political agendas, with left wing parties generally supporting environmental and green policies to a greater extent than right wing parties and environmental quality benefit from green parties being in leadership positions (Bernauer & Koubi, 2009). Differences in political agendas may influence the implementation and adoption of policies aimed at broader overarching goals. It may also influence the implementation of LEZs and, in turn, environmental outcomes in municipalities. The implementation strategies are often divided between command and control vs market oriented or goal based.

3.3.1 Command and control

The core categoristics of the command and control strategies works similar to the top down approach with rigid policies and mandates. The command and control strategy is often supported by penalties and fines to ensure compliance, rather than incentives to encourage adoption of the policy (Junquera & Del Brío, 2016). Left wing parties are more likely to employ this strategy to promote social equity, environmental sustainability, or “smart growth,” whereas right wing parties more frequently support the use of command and control approaches to maintain control through law and order measures (Fainstein & Novy, 2023). The command and control strategy is implemented in alignment with party objectives, and a degree of regulatory oversight is required to ensure that these policies are adhered to.

3.3.2 Market oriented strategy

The market oriented strategy places greater emphasis on fostering a competitive city in which privatization is promoted to attract investors and tourists; this represents a shift away from more traditional, government centered approaches focused on the public service model (Warner, 2012; Harvey, 1989). Left wing strategists sometimes use the market oriented strategy to promote private investment in social equity initiatives, affordable housing projects, and walkable cities aimed at reducing vehicle use, as well as public-private partnerships; this is intended to advance equality and sustainability (Peck & Tickell, 2002). The right wing emphasizes the potential opportunities, development, and investment benefits that privatization may generate, as well as the reduction of state involvement in sectors that can be privatized (Brenner & Theodore, 2005).

The command and control strategy is more favoured by the left wing, whereas the right wing to a greater extent tends to favour the market oriented strategy (Heywood, 2003), which in Swedish municipalities will potentially influence the implementation of certain policies (Knill et al., 2010).

4. Data and Method

This study is based on data from Statistiska centralbyrån (SCB), which collects data annually in Sweden at both local and national levels. The analysis is based primarily on data relating to passenger cars, which have been systematically collected and updated on an annual basis since 2006 along with data on nitrogen dioxide. SCBs goal is to provide statistics for decision makers, researchers and

to create a statistical foundation, and is endorsed by the Swedish government (SCB, n.d). Further, SCB provides the official statistics and is responsible for keeping and distributing the statistics, and creating a working system for the statistics and accessibility of the statistics.

4.1 Data

The following regions and their (code) are the primary research focus area that has been used and analysed in the study: Sweden (00), Uppsala (0380), Helsingborg (1283), and Umeå (2480). These cities were chosen due to when the low emission zones were implemented. Adding to that enough data was found. Stockholm, Gothenburg and Malmö were not selected because the implementation of low emission zones was done prior to 2000 and the lack of data prior to implementation makes it hard to analyse the pre-treatment effects, as well as the up to date interest. Mölndal was removed from the analysis even though the implementation was done more recent (2010), due to the closeness to Gothenburg and risk of spillover effects. Following this, the cities chosen for this research are also of various geographic locations. This is interesting because closeness to other cities, where it is located in the country, can have an effect on the policy implementation and effectiveness.

4.1.1 Variables

The variables utilized for the analysis were obtained from SCB and the Swedish Meteorological and Hydrological Institute (SMHI). The main variables in the research are Nitrogen dioxide levels, the total number of cars per 1000 inhabitants, cars which are owned by private persons and number of heavy lorries in use. The reasoning behind the variables in this study lies in all of them

having an impact on the environment and air quality. Also, the policy is directed towards the chosen variables and for that reason works well to research to contribute to understanding the low emission zone policy.

4.1.2 Nitrogen dioxide

To measure the effect on emissions the nitrogen dioxide (NO2) levels in the city are used. The air quality is heavily affected by NO2, and vehicles are a primary factor to NO2 emissions, and those pollutants have a great effect on both the environment and human health (Matthaios et al., 2019; Carslaw et al., 2018). To lower the emissions of NO2 is crucial to reach and uphold the public health standards, the lowering of emissions is urgent, as it is critical to have healthy and sustainable living environments (Taha et al., 2025). The NO2 variable is chosen because there is data on this emission, heavily caused by road vehicles and an important factor on both environmental sustainability but also public health.

4.1.3 Total number of cars and cars per capita

The higher number of cars in a place, the higher the risk of pollutants. The number of cars influence the emissions and in turn the environment, the air quality and the inhabitants well being. Cars per capita is used to examine how the policy primarily influences behavioural patterns and outcomes. Adding to this, the total vehicle fleet has a significant effect on air pollution and air quality in a city, particularly affecting people living close to roads, which is often observed in urban areas (Zhang and Batterman, 2013). Having both the total number of cars and privately owned cars per capita provides two layers of analysis and is therefore chosen. The two measures complement each other. One

focuses on overall change, whereas the other provides a deeper understanding of behavioural change.

4.1.4 Heavy lorries

The variable for heavy lorries was decided upon due to the fact that the LEZs primarily focuses on heavy traffic (Trafikverket, 2024). That makes it interesting to examine the impact of the policy on these types of vehicles, specifically heavy lorries. This variable is used to broaden the understanding of NO2, which serves as a proxy for emissions and has a direct link to air quality. Cars represent a more individual level influence on inhabitants, whereas heavy lorries function as a proxy more closely associated with industry, as they are primarily used for transport rather than day to day mobility. It is therefore interesting to assess whether LEZs have an impact on this category.

4.2 Methodology

This study has used ArcGIS as a tool to create maps based on quantitative statistics, in combination with Microsoft Excel, STATA and R for data analysis as well as the creation of diagrams. This combination was chosen to provide a clear structure for analysis and the maps give clear visuals on what it looks like mapped out.

4.2.1 Descriptive analysis

First, a descriptive analysis was conducted. This method was used to provide a clear visual understanding of the subject. By examining regions in Sweden and how they differ, this approach offers valuable insight into both current

conditions and historical developments, enabling the identification of trends over time and an assessment of how a policy can influence a specific place and its surroundings, without necessarily following the same pattern across all of Sweden (Stone et al., 2012). The key purpose of descriptive analysis is to illustrate measures of central tendency and provide clear, interpretable insights.

To create maps for the descriptive analysis ArcGIS was used, to provide the symbology on maps to visualise the descriptive statistics. For this part of the analysis, all of Sweden was used as a control group. The maps were classified using the Natural Breaks method, as this technique identifies natural groupings within the data and minimizes variation within classes while maximizing variation between them. As a result, it provides a more accurate representation of the spatial distribution of vehicle numbers and enhances the interpretation of geographical patterns in car ownership.

Table 1. Variables used in the descriptive analysis.

Variables	**Regions**	**Year**	**Source**
Total amount of passenger cars in use	290	2010-2020	SCB
Petrol	290	2010-2020	SCB
Electric	290	2010-2020	SCB
Diesel	290	2010-2020	SCB

It should be noted that Sweden's population exhibited continuous growth throughout the study period (see Figure 16 in Appendix). From 2006 to 2020, the population increased each year (SCB, n.d.). Demographic developments are

relevant to the interpretation of the results, as substantial changes in population size could influence both vehicle ownership trends and associated emission levels. Consequently, population growth was taken into consideration when selecting the classification method for the descriptive maps, contributing to the choice of Natural Breaks, which provides an effective representation of the underlying distribution of the data.

4.2.2 Event study

Then, an event study analysis was conducted to compare nitrogen dioxide emissions (the dependent variable) in order to assess the impact of the implementation of LEZs. The LEZs are implemented to lower emissions, NO2 emissions are heavily dependent on vehicles (Carslaw et al., 2018), and therefore decided to be used in the difference in differences (event study) analysis. NO2 emissions are primarily caused by diesel engines, and therefore the analysis of fuel type is of particular importance. In urban areas, road vehicles are a primary source of NO2 emissions, contributing to reduced air quality (Gariazzo et al., 2020), which is an issue that LEZs aim to address. For NO2 emissions the measurements are done in the city central of all cities (SMHI, n.d.), which suits the research as the LEZs are also implemented in the city central.

For the initial event study model, average NO2 concentrations were examined as the dependent variable. The cities of Umeå and Helsingborg were analyzed over the period 2006-2020. Uppsala was excluded from the nitrogen dioxide event study due to lack of data. The data for this analysis was downloaded from SMHI. Due to the staggered implementation of LEZs in 2014 and 2008, respectively, a staggered event study model was employed. To evaluate the

impact of LEZs on nitrogen dioxide concentrations, Skellefteå and Jönköping were selected as control cities for comparative analysis. When collecting data for Skellefteå, it was found that it had multiple values for the years 2006-2010, an average value based on those numbers was therefore calculated. In all event studies the three cities were used except for this one.

Further, an event study was made for passenger cars in use per 1000 inhabitants that are privately owned. The data for this analysis was downloaded from SCB. For this test Uppsala was added for the group with implemented policy and Västerås as control city, to now have six cities, three with implementation and three as control group. This was also a staggered event study. Vehicle ownership was normalized as the number of cars per 1000 inhabitants rather than the total count, as the latter is intrinsically confounded by variation in municipal population size. To further visualize the comparison in the total number of cars per 1000 inhabitants, cars which are owned by a private person graph were made in Microsoft Excel in combination with a calculation in the difference between each city with the implementation and the average of the total of Sweden.

To calculate the year to year differences, the values were subtracted from those of the preceding year to determine the magnitude and direction of change (positive or negative) and they were produced in Microsoft Excel. This was carried out for the three selected municipalities with LEZs and three comparable municipalities without LEZs. The results were also compared with Sweden as a whole, which served as a benchmark.

Furthermore, statistical analysis was employed to evaluate disparities between cities that implemented the policy and those that did not. An event study framework was employed to mitigate selection bias by isolating the effects of the policy within the treatment cities, utilizing a control group to account for temporal trends and counterfactual outcomes (Sasabuchi, 2021). The difference in difference method is highly relevant when evaluating policies, as it is one of the most used in these types of analysis, and also requires a relatively low volume of data (Jiménes & Perdiguero, 2025). Since this research focused on evaluating the LEZ policy, the event study method fitted very well with the aim and contributed to understanding the function of the policy.

Since environmental zone 1 in Sweden primarily targets heavy vehicles (Transportstyrelsen, 2024), another event study was made to test how the policy affected the number of heavy lorries in each municipality. In this test, all six cities that were used in the cars in use per 1000 inhabitants were used for this analysis. The data for this analysis was downloaded from SCB.

For the event study, three cities comparable in size, road infrastructure, and geographical location were selected to provide an appropriate control group. Along with the criteria, enough data had to be found to be eligible for the selection. The cities chosen to be the control group were Skellefteå, Jönköping and Västerås.

Table 2. Variables used in the event study and its amount of observations

Variable	Observations	Source
Nitrogen dioxide (NO2)	60	SMHI
Privately owned cars per capita	90	SCB
Heavy lorries	90	SCB

4.3 Model specification

The event study is conducted in three separated equations, each with one dependent variable i.e. Nitrogen dioxide, cars per 1000 inhabitants and heavy lorries. The data was divided into two groups: treated and untreated. This distinction was made to separate cities with LEZs from those without, to test whether the trend differs between municipalities with the policy implemented compared to others.

The empirical analysis utilized a staggered event study/event study framework using the xthdidregress aipw in Stata and corresponding code in Rstudio as estimator. This approach was selected due to the fact that it allows treatment effects to vary over time and across municipalities while accounting for differences in treatment timing (i.e. staggered modelling). Municipalities that implemented the LEZs function as the treatment group, whereas municipalities that have not implemented LEZs serve as the control group.

The baseline specification can be expressed as:

$$Yit=\alpha+\beta Dit+\gamma i+\delta t+Xit\theta+\varepsilon it$$

Where Yit represents the outcome variable for municipality (i) in year (t). Dit is a binary treatment indicator equal to one after the implementation of LEZs, showing that it is the treatment group. γi refers to municipality fixed effects, and δt captures year fixed effects. Xit represents a vector for control variables, while εit being the error term (Stata, 2025, p.3; Marcus and Sant'Anna, 2021).

The augmented inverse probability weighting (aipw) estimator was employed due to its double robust properties (Stata, 2025, pp.1-5; Drukker, n.d.). The estimator combines outcome regression with inverse probability weighting and produces consistent estimates if either the outcome model or the treatment assignment model is correctly specified. Double robustness reduces sensitivity to misspecification in either the treatment or outcome model specification, thereby increasing the robustness and reliability of the estimated treatment effect.

To evaluate the dynamic effects of low emission zone implementation, event time indicators were included to estimate treatment effects before and after policy adoption. This allows for the assessment of both anticipatory effects and the evolution of policy impacts over time (Marcus & Sant'Anna, 2021). The event study/event study specification also provides a statistical and visual analysis of the parallel trends assumptions by comparing pre and post treatment coefficients between the treated and control municipalities (Clark and Tapia-Schythe, 2021; Marcus and Sant'Anna, 2021). Standard errors were clustered at the municipality level, to account for autocorrelation within municipalities over time, both for the treated and control group.

A key identifying assumption in the event study/event study framework is the parallel trend assumption (Stata, 2025, pp. 3-6; Marcus and Sant'Anna, 2021), this assumption requires that, in the absence of treatment, treated and control municipalities would have experienced similar trends in the outcome variable over time. In more precise terms, any systematic differences between the groups must have been a constant during a longer period of time, along with not being correlated to the timing of said implemented policy (Marcus and Sant'Anna, 2021; Callaway and Sant'Anna, 2021). Any systematic differences between treated and control municipalities in the pre-treatment period are not caused by LEZ implementation, as the policy had not yet been adopted.

To assess the validation of this assumption an event study specification can be used. The event study allows for the estimation of dynamic treatment effects across treatments periods before and after implementation. Although the parallel trends assumption cannot be directly tested, an event study specification can be used to assess its plausibility. The event study estimates treatment effects for periods before and after implementation, allowing researchers to examine whether treated and control municipalities exhibited similar pre-treatment trends. Pre-treatment coefficients that are statistically insignificant and close to zero provide evidence consistent with the parallel trends assumption, whereas significant pre-treatment coefficients may indicate potential violations.

4.4 Ethical considerations

This paper has adhered to the ethical principles for good research practice established by the Swedish Research Council (Vetenskapsrådet) (2002). These

principles involve the careful handling of data, the respectful treatment of participants, and the transparent reporting of results to ensure honesty and integrity in research (Vetenskapsrådet, 2002). All data used in this thesis are publicly available at the regional level from Statistics Sweden and Swedish Meteorological and Hydrological Institute. Therefore, no specific restrictions were applied during data collection. The ethical considerations focused on informed consent and the eligibility to use the data, which both these sources are set out to provide.

Considerable emphasis was placed on ensuring the transparency, rigor, and replicability of the research conducted in this study. To support future replication and independent assessment, comprehensive explanations and justifications of the methodological framework and variable selection are presented. Furthermore, the dataset utilized in the analysis is thoroughly documented and is publically available. As the dataset is publicly available and intended for public use, further examination of legal and ethical considerations related to data access and utilization was considered beyond the scope of this study. The research does not include nor focus on religion, gender, sexuality or disability. For this reason, this study also poses minimal risks for participants. Continuing, there are also minimal risks for the Swedish inhabitants in any of the municipalities as there are a lot of people living there and the result is generalized over the municipality which does protect the specific individuals and it makes it hard to impossible to identify certain trends to a certain person.

4.5 Limitations

There is a very limited amount of data when analysing only three cities that have implemented the policy. The reason for only analysing three of eight cities

in Sweden that have implemented LEZs was due to when the implementation was carried out. This chosen approach was adopted because four of the five cities implemented low emission zones at an earlier stage, thereby ensuring the availability of a sufficiently long observation period before and after implementation. This temporal coverage is essential for conducting a reliable assessment of policy effects. Therefore, an analysis of the impact of the policy would be very difficult for the cities with earlier implementation. One city was also excluded from the analysis due to its geographic proximity to an existing treatment site with a long standing policy implementation; this proximity introduces a risk of spillover effects, which could bias the estimation of the treatment effect.

The limited number of observations in an event study creates a risk that the results are statistically significant but is in reality just “noise” by under/overestimating the standard errors (MacKinnon & Webb, 2020), this in turn makes the test sensitive to outliers. Having a small dataset with outliers has the risk of one outlier averaging out certain trends and affecting the total outcome to a higher extent than with a larger dataset. The generalizability also becomes lower, as the test only analyzes a few cities it can potentially show trends for those that do not occur on a larger scale, even if all the analyzed cities follow similar trends. The result is not as robust as it would be with a larger dataset.

5. Results

In this section the results are presented, first the maps will be presented to provide a visual understanding over the study area, along with diagrams

representing the descriptive data in the maps. This will provide a foundation for the results. The map represents the total number of cars and fuel type. Next, the results from the main event study analysis will be presented. This model will represent the effects of the LEZs implementation, to further understand how emission levels, as well as the number of cars and heavy lorries, have changed before and after the implementation of LEZs.

5.1 Descriptive findings

The descriptive findings provide an understanding of geographical trends in Sweden, since the main analysis is carried out for cities spread out in the country, the maps will give some additional information. The maps presented will be from 2010, 2014 and 2018, because they are all years for elections, and in 2010, one region has implemented the LEZs and the others has not.

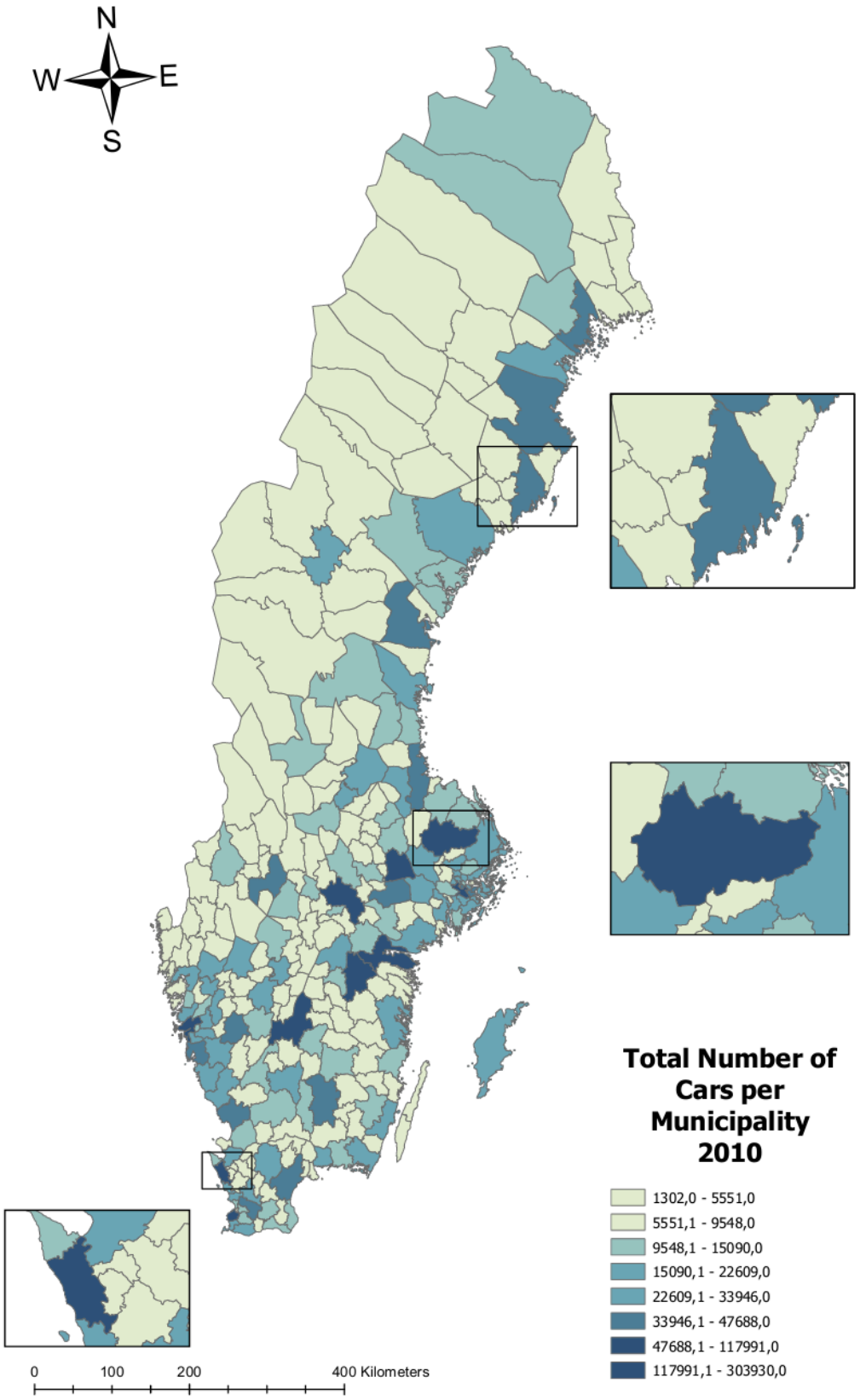


Figure 1. Total number of cars in Municipalities in Sweden 2010

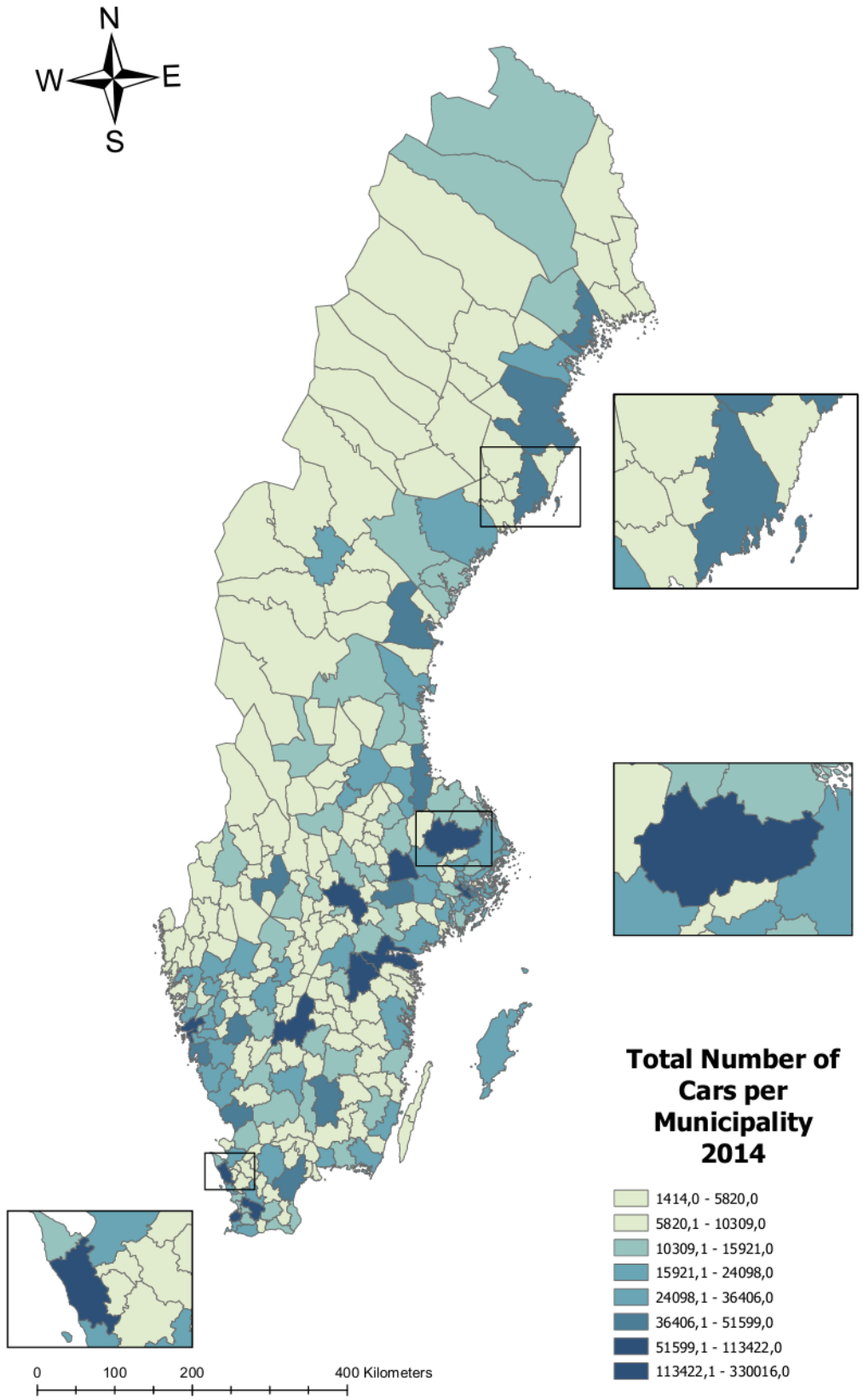


Figure 2. Total number of cars in Municipalities in Sweden 2014

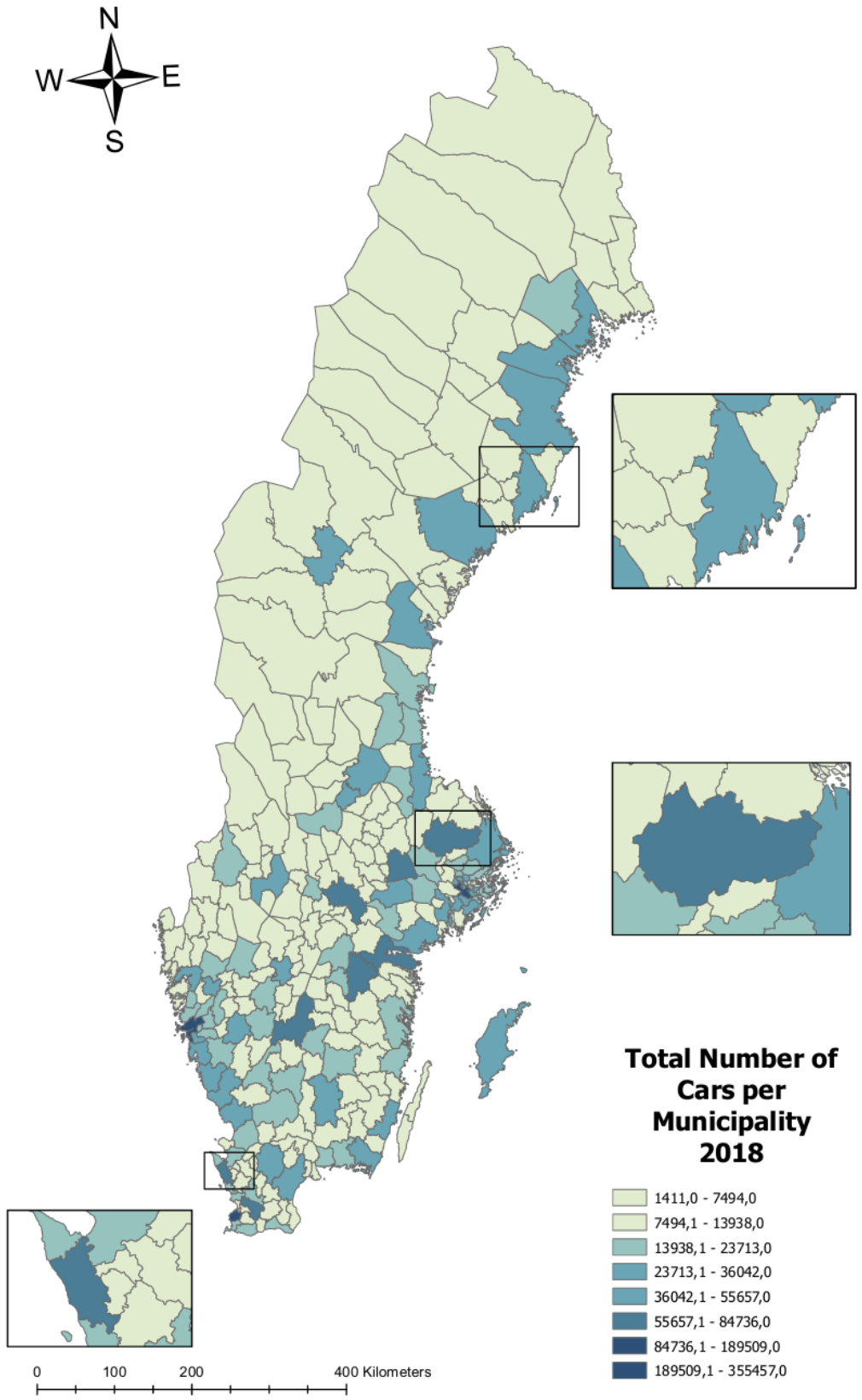


Figure 3. Total number of cars in Municipalities in Sweden 2018

The maps indicate that cities shift from dark blue toward lighter shades, suggesting a reduction in the total number of cars. However, this trend is not limited to the three cities where LEZs have been implemented. Rather, it represents a broader pattern observed across the entire study area. It can also be observed that the largest cities when it comes to population tend to have the highest overall number of cars, and despite being large to its size, northern municipalities remain the lowest in total number of cars. Consequently, the observed differences may reflect an imbalance in population size rather than substantive variation in vehicle ownership. The maps do also have different bracket intervals, in 2010 the highest dark blue bracket starts at almost 118 000, whereas in 2018 it starts at a bit over 189 500, this suggests that the total fleet of vehicles has increased, especially in urban centers.

There is also a pattern of darker blue shading in the areas surrounding the larger cities, indicating that the number of cars is likely influenced by commuting patterns. This suggests that the phenomenon is not confined to urban centers alone, but rather reflects a broader regional dynamic. The maps also insinuate that the car ownership is concentrated in areas with the best public transport networks (e.g. Stockholm, Gothenburg and Malmö).

Umeå has experienced a slight increase in the total number of cars. However, the map presents absolute values and therefore does not account for population differences between municipalities. As a result, some areas may display elevated numbers simply due to their larger population size rather than a higher level of vehicle ownership relative to the population. Specifically, Umeå has a substantial volume of through traffic; due to the absence of a fully completed bypass, a significant share of vehicles passes directly through the city. As a result, the actual flow of road traffic may be higher than what is reflected in the

mapped data. The implementation of the LEZs is to regulate these types of problems.

Uppsala is likely to function as a commuter hub, where the number of cars reflects flows of people residing in Uppsala but working in Stockholm, and vice versa. Helsingborg, by contrast, exhibits the most stable trend among the three cities. This stability is strongly influenced by its geographic location near Denmark, making it a transit point for through-traffic. The implementation of LEZs can therefore be understood as a measure to reduce transport-related traffic associated with the port. Additionally, the relatively stable trend may be explained by the early introduction of LEZs in Helsingborg, which were implemented as early as 2008.

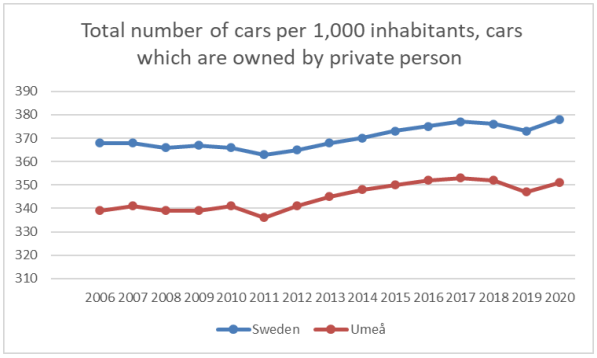


Figure 4. Comparison in privately owned cars between Umeå and Sweden

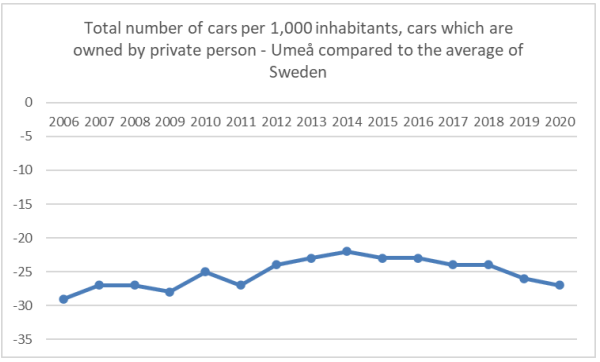


Figure 5. Difference in private ownership of cars in Umeå and Sweden

Figure 4 explains that there is a very parallel trend between car ownership in Sweden and Umeå, however, Figure 5 shows that Umeå has generally had a lower amount of cars per 1000 inhabitants compared to the average in Sweden. One reason for this could be the large number of students in Umeå. However, these trends allow Umeå to be a well suited city to implement LEZs as the population is already trending toward a lower use per capita. The trends observed in Figures 4 and 5 are supported by Figures 1-3, as the color scale remains relatively stable across them. Examining the areas surrounding Umeå in

Figures 1-3, these exhibit consistently lower numbers of cars than the city itself. This suggests that Umeås traffic and air quality challenges are primarily driven by internal factors or by transit flows. Consequently, the implementation of LEZs may not be aimed at addressing broader regional trends, but rather at mitigating the effects of Umeå's role as a transit corridor between southern and northern Sweden. Due to its geographical location Umeå also fights the challenge with temperature, as cars need to be heated or releasing more emissions due to cold starts as well as challenges with electric cars, this could also force people to use public transport and affect the lower ownership trend in Umeå. Moreover, Umeå being a student city does probably also influence the number of cars per capita. The number of students could therefore be important to acknowledge and understand that it may influence the trends.

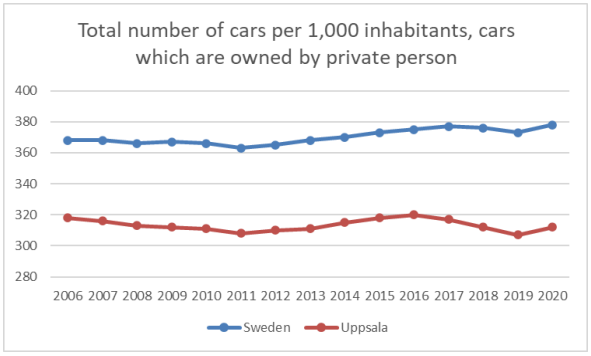


Figure 6. Comparison in privately owned cars between Uppsala and Sweden

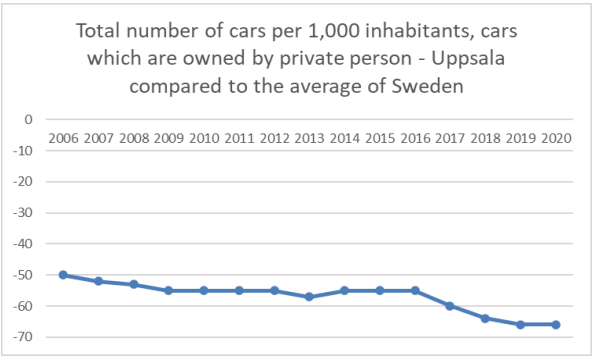


Figure 7. Difference in private ownership of cars in Uppsala and Sweden

In Figures 6 and 7, the trend for Uppsala is noteworthy, as it shows a widening gap between Uppsala and the national average for Sweden, with the difference becoming increasingly pronounced over time. This is particularly notable given that the maps in Figures 1-3 depict Uppsala as having a high volume of cars, whereas Figures 6 and 7 indicate a divergence from the broader national trend in Sweden. These trends suggest that Uppsala has a high urban planning focus on bikes, many students and an effective city planning in reducing car usage. This in turn means that even though the cars per capita is low, the total number of

cars are high and the LEZs can reduce traffic even more and the transport of goods through Uppsala can be lowered. Uppsala having a low car per capita ratio makes the LEZs primarily target surrounding areas rather than Uppsala residents.

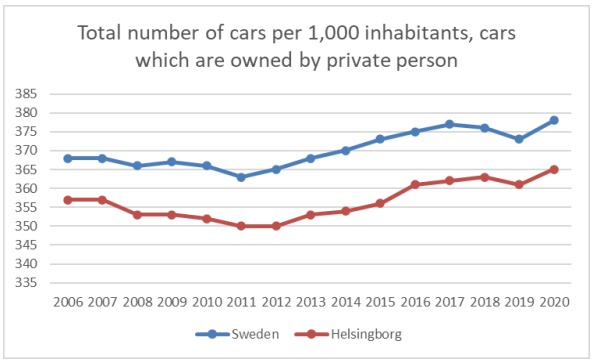


Figure 8 Comparison in privately owned cars between Helsingborg and Sweden

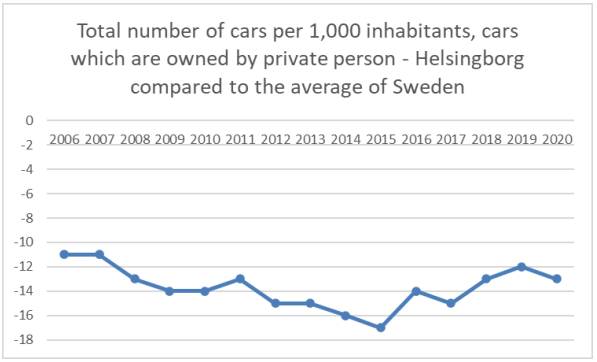


Figure 9. Difference in private ownership of cars in Helsingborg and Sweden

Helsingborgs trend is very similar to the national trend of Sweden, both in fluctuation and in numbers. This indicates that Helsingborg is more influenced by national factors (e.g. fuel prices, national policies and taxes), and less affected by regional regulations and planning. Due to its location and closeness to Denmark and natural hub to Europe from Sweden the trend being similar to Sweden it may not tell the full story, as heavy lorries in cities close to borders can worsen the air quality. Figures 1-3 indicate a high total number of cars, whereas Figures 8 and 9 display a pattern that is relatively close to the national average. This suggests that the number of cars is not solely determined by the municipality itself, but is also influenced by spillover effects from surrounding areas. Such patterns likely reflect strong regional flows, both within Sweden and internationally toward Denmark.

These three cities have all implemented LEZs despite being very different in many aspects, as figure 1-9 explains. Umeå being a regional hub for a larger geographical area, Uppsala being close to the capital with a widening trend

towards the national trend and Helsingborg being a city important to national transports. The cities all work with LEZs to improve air quality, using the policy to fight different challenges.

5.2 Event study analysis

Figure 10 shows us that there is a drop two years (-2) prior to the implementation being done as well as the direct effect of the implementation (0) is higher NO2 emissions. The relatively high value at year 0 may be explained by the fact that the policy does not produce an immediate effect, but rather a gradual impact over time. Around five years after implementation, the reduction in emissions becomes strong and consistent, persisting until approximately year 12. This pattern may be interpreted as a delayed treatment effect of the policy. However, despite the initial lag, Figure 10 indicates a subsequent positive trend after a few years, which is also statistically significant. The overall conclusion from these results is that the implementation of LEZs appears to reduce NO2 emissions in the long run. However, the pre-treatment dynamics are not clearly defined, and the results prior to implementation show some inconsistencies. As a result, the findings leave several questions unanswered, and the overall conclusions remain uncertain due to limitations in the robustness of the data.

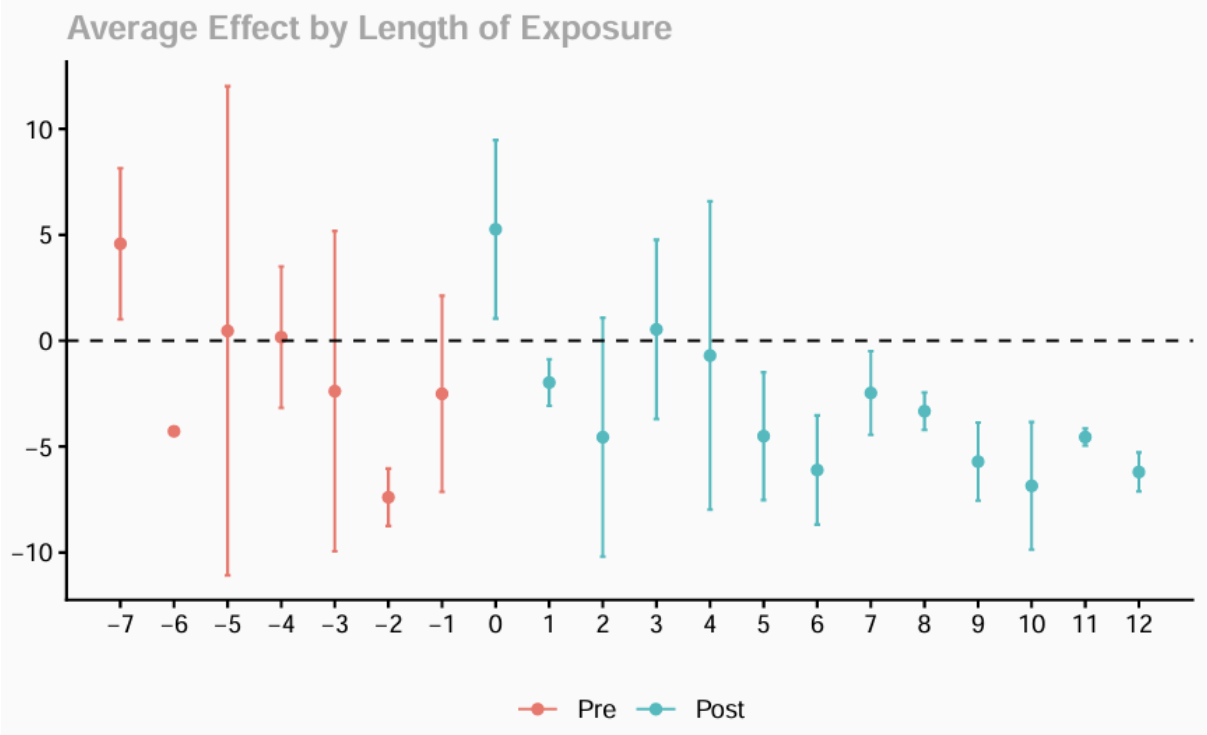


Figure 10. Average effect on Nitrogen Dioxide emissions by length of exposure to Low emission zones

Further examination of Figure 11 provides supportive evidence for the parallel trends assumption, as there are no strong pre-implementation trends and the estimated effects are not statistically significant prior to the policy's introduction. Figure 11 shows evidence of direct impact of the policy, with clear and significant reduction in privately owned cars per 1000 inhabitants. From the point of implementation onward, Figure 11 shows a negative, statistically significant, and increasing trend, indicating that the longer the policy has been in place, the greater its impact. The increasing effect may potentially be attributed to gradual adjustments over time. Figure 11 further explains that LEZs do not only have an immediate or temporary effect, instead it grows over time, that is a sign of persistence of the policy. The long term effects show tendencies of behavioural changes, and behavioural adaption that makes the policy strengthening over time. Overall, figure 11 provides strong evidence that LEZs lower the number of privately owned cars per capita, both a direct impact

of the LEZ but the effect also gets stronger over time and creates a long term effect. This is also evident in Table 3 in appendix, which shows that there is no significance before implementation, showing that there are no trends prior to implementation and the policy itself has therefore a strong impact.

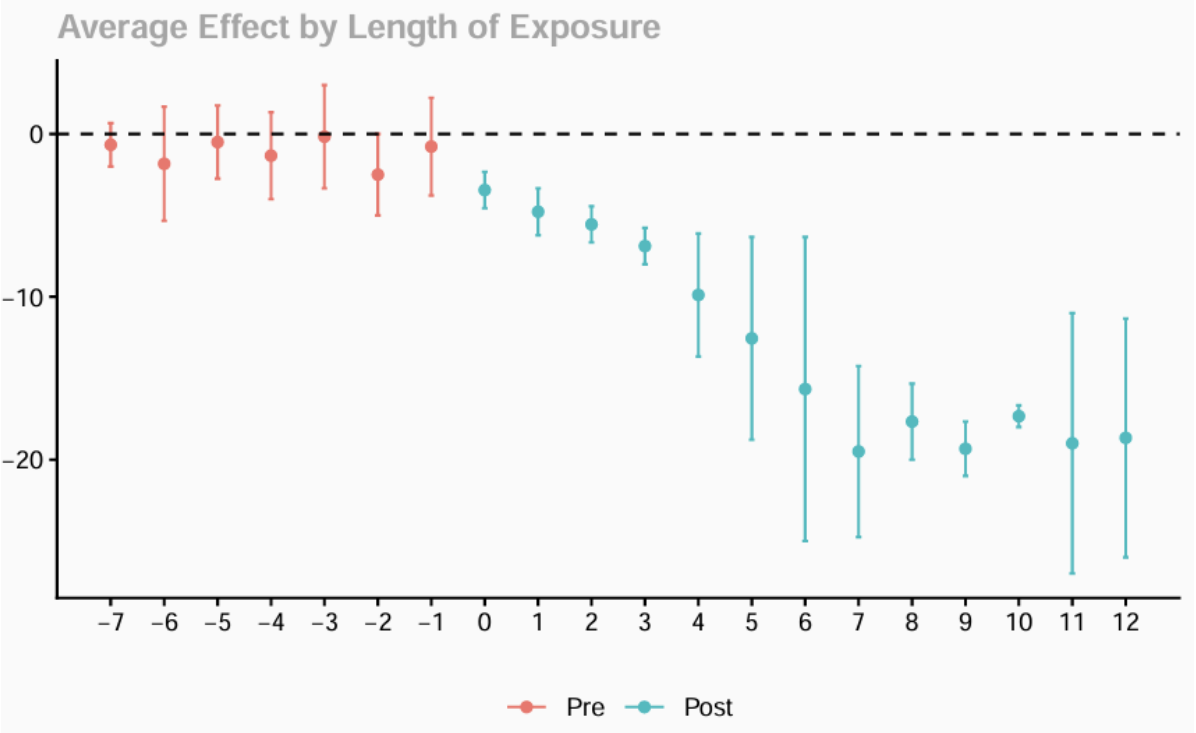


Figure 11. Average effect on privately owned cars per 1000 inhabitants by length of exposure to Low emission zones

In figure 12 no real evidence could be seen, as all the years prior as well as post are insignificant and follow the same trends. This could be due to not enough variation in the data. In conclusion figure 12 is inconclusive, it has no statistical power and the estimates are highly imprecise. The conclusion is that no meaningful inference can be drawn due to the wide confidence intervals. In turn, figure 12 gives us no further information on the direct effect LEZs has on the number of heavy lorries.

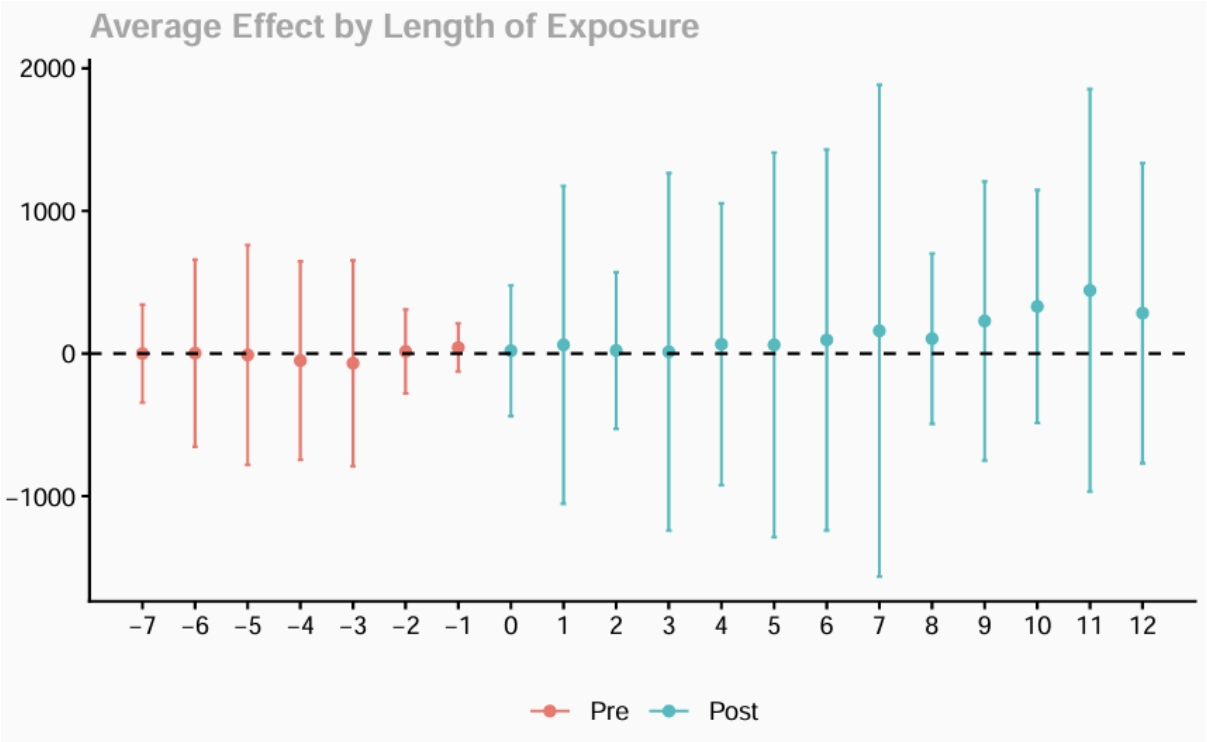


Figure 12. Average effect on heavy lorries by length of exposure to Low emission zones

Overall, the event study gave clear evidence for privately owned cars, tendencies on NO2 emissions and no statistical evidence for heavy lorries. Further, LEZs have a negative effect on both NO2 and primarily privately owned cars, which is the goal of LEZs, as the aim is to promote more sustainable means of transportation (Transportstyrelsen, 2024). By reducing the number of cars and associated emissions, air quality is expected to improve, contributing to more sustainable urban environments. These findings also provide evidence that LEZs have the potential to promote environmental sustainability.

Further understanding of NO2 emissions could be found in newly registered cars. As diesel cars are a high NO2 polluter (Li et al., 2018), the emissions are affected by the type of fuel used by vehicles. In figure 13 it is clear that along with the lowered number of cars in the studied areas shown in figure 11, the number of diesel cars are also reduced. In all three areas, newly registered cars

are seen having reduced levels of diesel cars. This is also a general trend nationwide. In the brackets in figure 13 it is clear that the total number of new cars with diesel fuel is lower, meaning that less cars with diesel fuel were registered in 2020 compared with 2010.

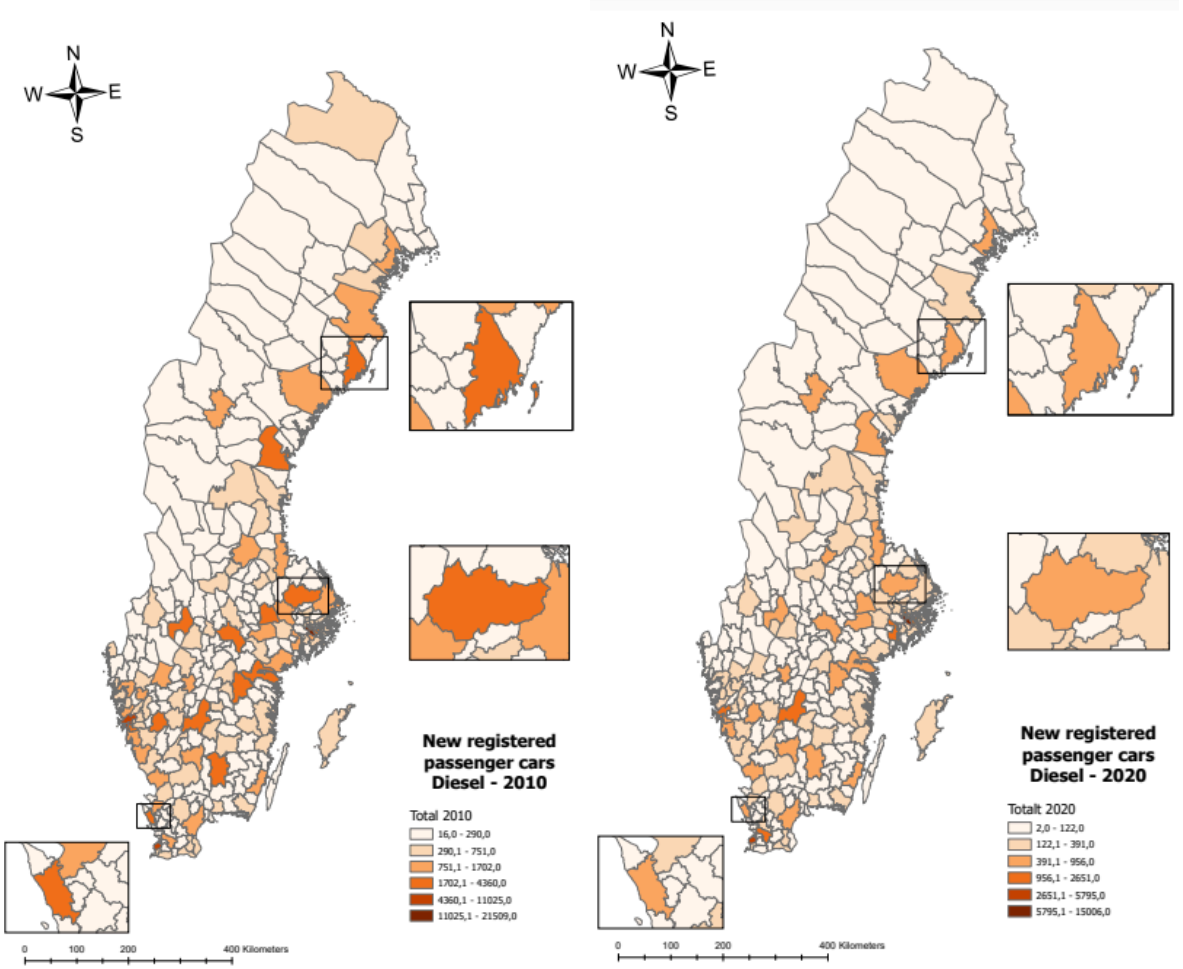


Figure 13. Newly registered diesel cars comparison between 2010 and 2020.

Figure 14 illustrates a huge growth in electric cars between 2010 and 2020, this aligns well with the goal of LEZs, as the growth of electric cars lowers emissions (Singh and Namrata, 2025). However, this growth occurs throughout Sweden, but is somewhat more pronounced in cities with LEZs. Helsingborg, which implemented the LEZ in 2008, is of particular interest when examining the map for 2010, as it records among the highest values at that time. Umeå is also an interesting case, as it is the city in northern Sweden in 2020 with the

highest growth since 2010, and the only city in northern Sweden that does not fall within the lowest bracket of electric cars. Although the overall ownership of electric cars in Sweden is higher in 2020, none of the studied cities fall within the lowest categories, indicating a growing interest in sustainable practices among the population.

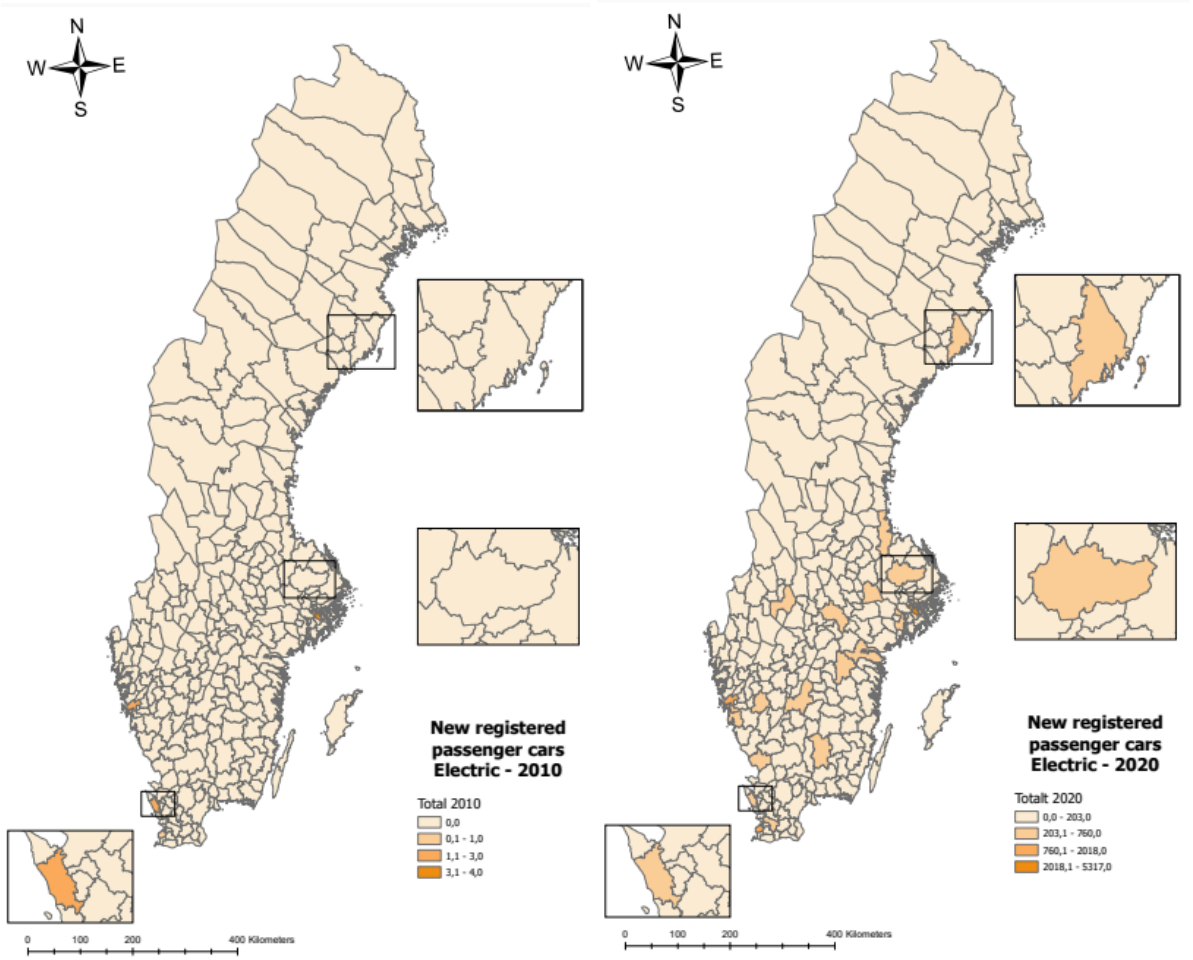


Figure 14. Newly registered electric cars comparison between 2010 and 2020.

In figure 15 the map illustrates a much more stable development from 2010 to 2020. As the difference in the categories as well as the development in Sweden has not grown a lot. It should be noted that the category is higher in 2020, indicating that the total number has increased. Thus, even if a municipality remains in the same category, its absolute number has still grown. This

development indicates that the growth is spread out portionally and mainly in the bigger cities.

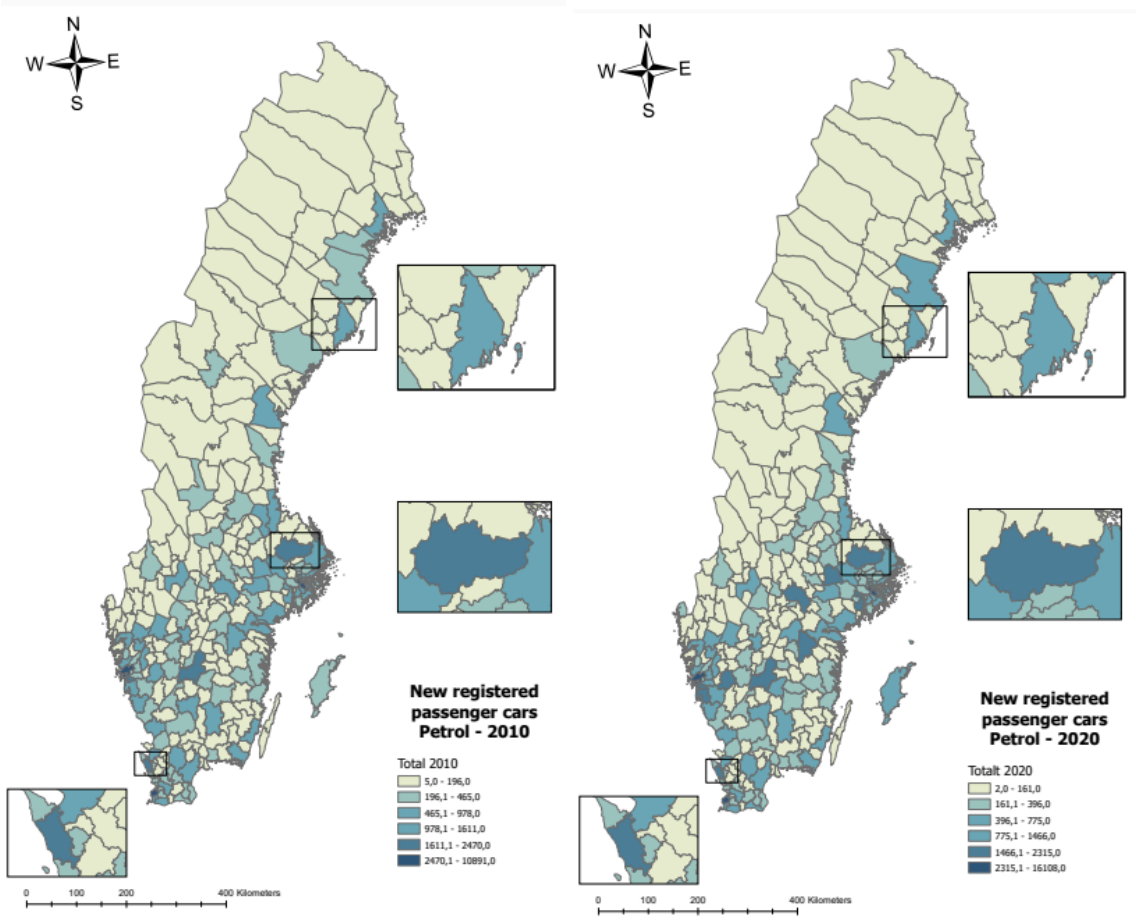


Figure 15. Newly registered petrol cars comparison between 2010 and 2020.

The stability in petrol is likely due to the fact it is acting as a safe choice. While electric car adoption has increased (figure 14) and diesel usage has declined (figure 13), concerns regarding reliability have contributed to a rise in hybrid vehicles, typically combining electric and petrol systems. Moreover, following the Volkswagen emissions scandal in 2015 (Jung & Park, 2017), choosing a hybrid or petrol car has often been perceived as the more straightforward option.

6. Discussion

This section interprets the results in relation to the theories outlined in the theoretical framework, with a focus on integrating these perspectives to develop a deeper understanding of the findings.

6.1 Are the low emission zone policy showing effectiveness?

The low emission zones primary target is to improve air quality, by reducing emissions and transport through cities (Trafikverket, 2024). The results section clearly demonstrates a reduction in NO2 emissions, which are a central component of air pollution, particularly in relation to road vehicles. Figure 10 illustrates tendencies of lowered NO2 emissions, which is one of the goals with NO2 emissions. However, these tendencies are not clear results. The tendency shows that the LEZ partly lowers the emissions, but it is not possible to fully evaluate the NO2 emissions as fully successful as the result is not robust enough. The LEZ does only have partial effectiveness on NO2 emissions, this could be due to different reasons. Potentially longer exposure to the policy along with more data create more robust data for future studies and a more complete understanding can be made. As mentioned earlier, both the NO2 measurements and the LEZ are implemented in the city centres. However, the results are presented for the municipality as a whole. This means that the findings may be somewhat skewed, but for the purposes of this study, the primary interest is how air quality changes in urban areas, as these are often the locations most affected within the municipalities.

When examining the number of cars, both the total number and cars per capita exhibit a negative trend, indicating that these figures are declining relative to the

national average. The LEZ has shown good results on the number of cars. Figure 11 illustrates that the LEZ has an immediate effect on cars per capita, with this effect strengthening over time as exposure to the policy increases. The risk of overly positive results that governments try to modify in some instances (Handoyo, 2024), are eliminated. As the results are based solely on observed values and the analysis is conducted using two different statistical tools, with the primary aim of testing the policy, no additional assumptions are imposed. Moreover, the number of cars is unlikely to be fixed, as it is a statistical measure that does not necessarily correspond to any specific indicator of national success. The strength of the data comes from it being tested in more than one city which enables more data to make it more robust and less limited, which could be the case with a low amount of data, but it also needs to be understood that more data will make the test more robust, hence more trustworthy (Seelarbokus, 2014). The test on cars per capita is however strong and the graph states that the policy has an effect on the number of cars and behavioural patterns.

Further looking at heavy lorries, the result is more difficult to analyse. As there is no clear pattern along with low to no significance, it is tough to make any conclusion whether the policy has an effect or not. The effectiveness of the policy is therefore hard to impossible to draw any conclusion from the data on heavy lorries. Still, it could be interesting to investigate this in future research to understand the impact LEZs have on this type of vehicle.

Further looking at the environmental indicators, the LEZs has generally had a functioning impact, by lowering the nitrogen dioxide levels which in turn will improve air quality (Matthaios et al., 2019; Carslaw et al., 2018). The lowered

emission levels is likely caused by a reduction in cars and heavy traffic. This along with better planned cities. As the measures of nitrogen dioxide are made in city centers in this research, better planning by redirecting heavy loads of traffic around the city instead of straight through can improve air quality and emission levels. However, as the tests state that the LEZ implementation has had a direct impact on both cars per capita along with lowered emissions it is possible to state that the policy has a direct correlation with the environmental indicators. Finally, the effectiveness based on the environmental indicators are positive and it is a functioning policy in that regard.

For the behavioral outcomes there is one hand clear indication on success, as the cars per capita in the regions with implemented LEZ has a clear effect already year one of exposure. This effect also strengthens through time, imposing behavioral changes and patterns for long term development. However, the results for heavy lorries remain inconclusive, and the associated behavioral effects cannot be established with the same degree of confidence as those observed for passenger cars (Smith & Stirling, 2007). This uncertainty may, in part, stem from limitations in data availability and coverage, which restrict the capacity to identify clear policy effects. Given the substantial contribution of heavy duty vehicles to transport related emissions, a more comprehensive understanding of how such vehicles respond to low emission zone policies remains an important area for future research, as the policy in this case mainly focuses on changes within the local region.

While heavy lorries are more influenced by work and corporate transport, as this is counted on total number of heavy lorries and the cars per capita is tested on privately owned level, the behavioral indicators and impact on the municipality

is tighter related to cars per capita rather than heavy lorries. These findings, combined with the likelihood that heavy duty traffic is more responsive to regional and national regulatory measures, may indicate that local policies exert only a limited influence on this sector. This, in turn, suggests that heavy duty traffic is predominantly governed by higher level policy frameworks rather than municipal policy interventions. Meaning that the implementation has a fulfilling outcome on the inhabitants on a private level but has not reached the same extent of success on corporations.

The technological outcomes are tougher to analyse. The pattern is obvious, more electric cars, less diesel cars and a stable development for petrol cars between 2010-2020. However, it is difficult to understand what impact the LEZ has had on this pattern. The research areas have been front runners for more environmental alternatives based on the maps, this is not significantly an effect of the LEZ but could potentially be the other way around. As these cities have been up to speed with environmental alternatives, it could be easier to implement LEZ in those regions. It could therefore be a correlation between fuel types (technological advancements) and LEZ, but the correlation is potentially intertwined and not causal, rather mutually influencing. It is also the reason why it is difficult to analyse the effectiveness of the policy on technological outcomes.

The indicators and outcomes provide a good foundation on evaluating the effectiveness of the policy, but it is important to not oversimplify the result (Mickwitz, 2003; Gudmundsson, 2003). There are factors that influence the policy effectiveness and it is therefore important to understand that the correlation sometimes has mutual effectiveness in some instances. The

outcomes do give us a general understanding of the effectiveness, which indicates that the policy is effective to a high extent, especially for behavioral patterns.

6.2 Does the policy implementation work well?

Mathison (2005) and Grindle (1980), enforces power to the actions surrounding the policy, not only if the outcomes are good but if the implementation works. Based on the result it primarily provides information on if the policy works or not, or to what extent it works. The interesting part is that the success of the policy has similar trends in all researched regions, this can potentially explain a working implementation in all three areas. Based on the results and the aim of the policy, it illustrates that the policy has been translated from policy into action in a functional manner. Since the policy focuses on air quality and lowering emissions, the implementation of the policy must have succeeded to some extent to yield reduced levels of nitrogen dioxide and thereby improved air quality. The results provide evidence that the policy has been carried out effectively, supporting the conclusion that its implementation is a functioning mechanism.

Sweden often works as a middle way between bottom up and top down governance, with an overarching goal, still large decision making on municipal level. Sweden has a nationwide aim to become more environmentally friendly, but the introduction of LEZ is made on local/municipal level. This creates a middle way between the overarching national goal aligning with local goals and initiatives. However, in this instance it works well, since the aim of the policy focuses on the same goals as Sweden in general. The regional initiatives in this

case LEZ, could grow to a larger level, from bottom up to where the government decides on expanding (top down) to more cities in Sweden if proven successful. For the current state of Sweden it makes implementation of LEZ easy for regions and municipalities as they can decide it on their own, this further enables possibilities to work closer to the policy and make implementation work for each current place. This in turn often makes it easier to make policies effective as they work with proactive implementation (Kaiser, 2020). The overarching goal does add administrative advantages, helps with creating policies that focus clearly on achieving goals and objectives (Li, 2023), uniforming the policies nationwide, to aim for the same objectives as well as creating a clear direction for the policy.

6.3 Urban politics on low emission zone

The research cities have through the last decades different results in the local election, with both the results tending to be a bit more left in Umeå and Uppsala with Helsingborg being more split in half between left and right (Valmyndigheten, 2025; SCB, n.d). This provides information on which municipalities tend to implement this policy, as local enforcement may influence a municipality's willingness to adopt low emission zones. All individuals are a part in creating the social patterns (Stoker, 1998), this can be reflected in the behavioral patterns, mainly illustrated by privately owned cars per capita. Breaking the LEZ rules often results in fines (Trafikverket, 2024), meaning the policy operates with control as its main regulating factor. At the same time the LEZ can be argued to be an incentive to drive more environmentally friendly vehicles, meaning that command is its main regulating factor. What can be seen is that the researched cities are trending more to the left in general during the tested years, and the results have been somewhat

successful. The implementation and the results seems to be influenced by the municipal leaders and willingness by the inhabitants, as the left side often are more focused on environmental sustainability compared to the right wing parties. This explains that what party has the political power affects both the implementation and effectiveness. However, since we do not have a municipality in this test that is clearly more right wing, it is hard to test for the political factor and should be considered for future studies. Especially since the tested cities are drifting in different directions the last few years. It would also be interesting to make an analysis on willingness to implement LEZ in municipalities based on self placement on the left to right scale.

Command and control measures strongly influence behavioral patterns, which in turn affect both environmental and technological outcomes. By changing behavioral patterns through command and control measures, changes occur in which cars are purchased and used, followed by changes in emissions. The command and control works on a sociological level where different choices are more accepted than others (Månsson, 2010), e.g. a vehicle that is more sustainable is more accepted by today's society. This aspect influences all of society, and trends towards sustainable development may accelerate with LEZ, as it signals which cars are considered more acceptable within the car industry. This is also reflected in the results, where longer exposure to the policy corresponds to a greater effect on the number of privately owned cars per capita. This may indicate a shift towards increased bicycle use and, consequently, fewer cars.

The behavioral patterns influence the market, with a higher willingness to go electric will force a higher production or higher price. This is illustrated in

figure 14, where the electric car fleet has grown substantially. With a population increasingly inclined to purchase electric vehicles, car manufacturers will produce more, resulting in greater competition among companies. A higher supply of electric vehicles may reduce prices and thereby enable more people to afford them. This makes the transition both more accessible and socially encouraged, combining command and control measures related to car ownership, framed and supported by LEZ. Furthermore, the car market in the researched municipalities is likely smaller compared to nationwide averages, as LEZs indirectly promote alternative means of transportation, thereby affecting the demand for cars. With lower demand and fewer cars in circulation, there is a high probability of reduced emissions, which is also illustrated in figure 10. In conclusion, the implementation of LEZs influences market supply and demand, as these regulations affect both technological and behavioral patterns. Based on the objectives of the policy, LEZs demonstrate positive outcomes in the market.

Adding to this, the population trend in Sweden has been stable and slightly growing each year. This development would, in principle, be expected to increase emissions. However, the concurrent decline in the number of cars per capita may offset this effect, as indicated by the observed downward trend in vehicle ownership relative to population size. It is therefore important to distinguish between the total number of registered vehicles and the number of vehicles per capita, as these indicators may exhibit divergent trends and lead to different interpretations. In the end, the goal of LEZ is to increase air quality, which less cars per capita and a change towards more electric from diesel tend to do based on the nitrogen dioxide levels.

6.4 Future Research

Future research should consider expanding the dataset to include all Swedish cities that have implemented low emission zones, thereby enabling a more comprehensive evaluation of their effectiveness. Additionally, a comparative study examining the impacts of low emission zones in the three largest and three smallest participating cities could offer further insights into the extent to which urban characteristics, such as population size and spatial structure, influence policy effectiveness.

For heavy traffic (i.e. heavy lorries) other vehicles could be added along with testing more regional or national policies and its effect on heavy traffic. Furthermore, future research should explore the impact of regional and national policy measures on heavy lorries and/or other forms of commercial heavy duty traffic. Given that the findings of this study indicate a limited or inconclusive effect of local policy interventions, examining policies implemented at higher levels of governance may provide further insight into the most effective mechanisms for regulating heavy duty transport and reducing its environmental impacts.

7. Conclusion

In this section the results of the aim and research question will be analysed, discussed and concluded. It will refer back to the three research questions in this thesis.

7.1 The implementation of low emission zones in Swedish cities influence on changes in car ownership

The results clearly illustrate that the researched municipalities with implemented LEZs consistently exhibit lower numbers of cars compared to national averages. Furthermore, the implementation of low emission zones appears to reduce cars per capita progressively with longer policy exposure. Can the result answer: To what extent are low emission zones in Swedish cities associated with changes in the total number of cars and cars per capita over time? The results illustrate that LEZs influence car ownership with robust effects and have the strongest and most robust impact on cars among the tested variables.

The results demonstrate the efficiency and effectiveness of the policy, highlighting its importance and providing justification for its implementation in additional areas to improve air quality and address environmental challenges that must be taken into consideration. Furthermore, the policy offers an effective means of reducing the number of cars on the road and, consequently, car related accidents. Even if the total size of the car fleet has grown, the rate of growth has slowed, and in the studied cities the growth speed has been reduced, alongside a shift in fuel types. The results indicate that the trend towards more sustainable vehicles has begun and strengthened, highlighting the influence of LEZs. In conclusion, the exact extent to which low emission zones are associated with changes in car ownership is difficult to determine, but it is evident that LEZs have a positive environmental, behavioural, and technological influence on the car fleet.

For future research, it would be interesting to go deeper into other policies or similar that potentially influence car usage. The results clearly show that the researched municipalities have lower numbers of cars and continue to decrease during exposure to LEZs. However, there may be factors that play a greater role than anticipated, even though the ATET test in figure 11 and table 2 yields significant results. So, for future studies it would be interesting to test which factors strengthen and weaken the power that LEZ has on car ownership according to the result in this research paper.

7.2 Low emission zones influence on the composition of vehicle fleets

It is evident that there have been changes in the vehicle fleet nationwide between 2010 and 2020. Further, how do low emission zones influence the composition of vehicle fleets, particularly the share of different fuel types? The electric vehicle fleet has grown, while the diesel fleet has declined, and petrol has remained relatively stable. The exact influence of low emission zones is in this case difficult to draw firm conclusions about. Although Umeå, Helsingborg, and Uppsala have been frontrunners, with early high numbers of electric vehicles and a reduction in the diesel fleet, this is not necessarily attributable to low emission zones. Although it may be difficult to analyse the influence of LEZs on the car fleet, such analysis provides important information for both government officials and individuals purchasing new cars, particularly regarding current trends and potential future policies.

It is also possible to understand the dual nature of LEZs by examining these trends. This suggests that implementation may be influenced both by a willingness to adapt to evolving policies and social trends among the

population, and by that, the municipal government's intention to act in the interests of its inhabitants. In conclusion, the exact effect of LEZs on the car fleet is not fully clear, although the information they provide is highly relevant and creates a foundation for understanding national trends in comparison to the study area. It also symbolises behavioural patterns to a large extent, which influence decision making at a higher level.

For future studies, it would be advisable to include hybrid vehicles, as this could address a gap in the existing research. Additionally, more in depth statistical analyses should be conducted rather than relying solely on descriptive conclusions. The analysis should ideally incorporate the direct influence of LEZs as well as other potential contributing factors.

7.3 Low emission zones affect on environmental outcomes, behavioral changes, and technological adaptation

The behavioural changes are largely reflected in changes in cars per capita, on which LEZs appear to have a significant impact. Similar trends can not really be observed for heavy lorries. However, this aspect is more difficult to analyse. Since the results are not sufficiently robust to be considered reliable, it is not possible to determine whether the impact is positive or negative. A potential explanation for this result is that heavy lorries vehicles are not primarily affected by municipal policy measures but are instead more strongly influenced by regulatory frameworks established at the regional or national level. For that reason, the behavioural impacts of LEZs appear to contribute to environmental improvements and a reduction in car use, particularly at the private level. On a broader scale, industrial trends are more difficult to analyse, as they are more

established and therefore harder to change. Ultimately, LEZs do appear to influence behavioural trends at the individual level.

Looking at the technological adaptation, the largest understanding is the long term changes in fuel type seen from 2010 to 2020. The impact of LEZs is not entirely clear. Technological advances have been more prominent in the researched cities, but whether this is a result of the LEZs is difficult to determine. The outcome of LEZ on technological advances needs further research to investigate in the future to determine the effect of the policy.

Furthermore, the environmental outcomes associated with LEZs show clear indications of reduced emissions, particularly regarding nitrogen dioxide, which is an air pollutant largely caused by road vehicles. LEZs are lowering the NO2 emissions and that in turn promotes better air quality. Furthermore, improved air quality and environmental benefits are observed in cities that have implemented LEZs. This is positive in terms of human sustainability, quality of life, and more environmentally friendly living. What has to be taken into consideration is that the fuel type may influence the emissions, and since it is not clear to what extent the LEZ influences fuel type, the emissions could also be affected by fuel type and not only the LEZ.

So, to what extent do low emission zones affect environmental outcomes, behavioral changes, and technological adaptation? In conclusion, LEZs have an impact on behavioural and environmental patterns, while their influence on technological advances is more difficult to determine. The analysed results

indicate that LEZs are effective and should be considered for implementation in more cities.

8. References

Bateman, I. J., Harwood, A. R., Watson, R. T., Abson, D. J., Andrews, B., Binner, A., Crowe, A., Day, B. H., Dugdale, S., Fezzi, C., Foden, J., Hadley, D., Haines-Young, R., Hulme, M., Kontoleon, A., Lovett, A. A., Munday, P., Pascual, U., & Termansen, M. (2013). Bringing ecosystem services into economic decision-making: Land use in the United Kingdom. Science, 341(6141), 45-50. https://doi.org/10.1126/science.1234379

Bernauer, T., & Koubi, V. (2009). Effects of political institutions on air quality. Ecological Economics, 68(5), 1355-1365. https://doi.org/10.1016/j.ecolecon.2008.09.003

Brenner, N., & Theodore, N. (2005). Neoliberalism and the urban condition. City, 9(1), 101-107. https://doi.org/10.1080/13604810500092106

Broster, S., & Terzano, K. (2025). A systematic review of the pollution and health impacts of low emission zones. Case Studies on Transport Policy, 19, 101340. https://doi.org/10.1016/j.cstp.2024.101340

Bruyneel, L., Cox, B., Stauffer, A., Vandenthoren, L., Fierens, F., Nawrot, T. S., & Horemans, C. (2025). Positive impact of the introduction of low-emission zones in Antwerp and Brussels on air quality, socio-economic disparities and health: A quasi-experimental study. Environment International, 199, 109515. https://doi.org/10.1016/j.envint.2025.109515

Bullock, H., Lavis, J. N., Wilson, M. G., Mulvale, G., & Miatello, A. (2021). Understanding the implementation of evidence-informed policies and practices

from a policy perspective: A critical interpretive synthesis. Implementation Science, 16, Article 82. https://doi.org/10.1186/s13012-021-01082-7

Carslaw, D., Farren, N., Vaughan, A., Drysdale, W., Young, S., & Lee, J. (2018). The diminishing importance of nitrogen dioxide emissions from road vehicle exhaust. Atmospheric Environment: X, 1, 100002. https://doi.org/10.1016/j.aeaoa.2018.100002

Callaway, B., & Sant'Anna, P. (2021). Difference-in-Differences with multiple time periods, Journal of Econometrics, Volume 225, Issue 2, Pages 200-230, ISSN 0304-4076, https://doi.org/10.1016/j.jeconom.2020.12.001

Clarke, D., & Tapia-Schythe, K. (2021). Implementing the panel event study. Stata Journal, 21(4), 853-884. https://doi.org/10.1177/1536867X211063144

Drukker, D. (n.d.). Double-robust treatment effects. Stata News, 29(1). https://www.stata.com/stata-news/news29-1/double-robust-treatment-effects/

Eisman, A. B., Kilbourne, A. M., Dopp, A. R., Saldana, L., & Eisenberg, D. (2020). Economic evaluation in implementation science: Making the business case for implementation strategies. Psychiatry Research, 283, 112433. https://doi.org/10.1016/j.psychres.2019.06.008

European Environment Agency. (n.d.). Indicators. https://www.eea.europa.eu/en/analysis/indicators

Fainstein, S., & Novy, J. (2023). Right-wing populism and urban planning. Journal of Urban Affairs. 47. 1-24. 10.1080/07352166.2023.2200950

Filgueiras, F., Palotti, P., & Testa, G. G. (2023). Complexing Governance Styles: Connecting Politics and Policy in Governance Theories. Sage Open, 13(1)

Flanagan, E., Malmqvist, E., Gustafsson, S., & Oudin, A. (2022). Estimated public health benefits of a low-emission zone in Malmö, Sweden. Environmental Research, 214(Part 4), Article 114124. https://doi.org/10.1016/j.envres.2022.114124

Gariazzo, C., Carlino, G., Silibello, C., Renzi, M., Finardi, S., Pepe, N., Radice, P., Forastiere, F., Michelozzi, P., Viegi, G., & Stafoggia, M. (2020). A multi-city air pollution population exposure study: Combined use of chemical-transport and random-forest models with dynamic population data. Science of the Total Environment, 724, 138102. https://doi.org/10.1016/j.scitotenv.2020.138102

Grindle, M. S. (1980). Policy content and context in implementation. In M. S. Grindle (Ed.), Politics and policy implementation in the Third World (pp. 3-34). Princeton University Press.

Gudmundsson, H. (2003). The Policy Use of Environmental Indicators-Learning from Evaluation Research. The Journal of Transdisciplinary Environmental Studies. 2.

Handoyo, S. (2024). Public governance and national environmental performance nexus: Evidence from cross-country studies, Heliyon, Volume 10, Issue 23, e40637, ISSN 2405-8440, https://doi.org/10.1016/j.heliyon.2024.e40637

Harvey, D. (1989). From managerialism to entrepreneurialism: The transformation in urban governance in late capitalism. Geografiska Annaler: Series B, Human Geography, 71(1), 3-17

Heyden, M., Fourné, S., Koene, B., Werkman, R., & Ansari, S. (2016). Rethinking "Top-Down" and "Bottom-Up" roles of top and middle managers in organizational change: Implications for employee support. Journal of Management Studies, 54. https://doi.org/10.1111/joms.12258

Heywood, A. (2003). Political ideologies : an introduction (3. ed.). Palgrave Macmillan.

Ioannou, K., Karasmanaki, E., & Tsantopoulos, G. (2025). Environmental Policy as a Tool for Sustainable Development. Sustainability, 17(8), 3674. https://doi.org/10.3390/su17083674

Jiménez, J.L., Perdiguero, J. (2025). Difference-in-Difference. In: Marciano, A., Ramello, G.B. (eds) Encyclopedia of Law and Economics. Springer, Cham. https://doi.org/10.1007/978-3-031-76422-6_664

Jung, J., & Park, S. (2017). Volkswagen's Diesel Emissions Scandal. Thunderbird International Business Review. 59. 10.1002/tie.21876.

Junquera, B., & Del Brío, J. Á. (2016). Preventive Command and Control Regulation: A Case Analysis. Sustainability, 8(1), 99. https://doi.org/10.3390/su8010099

Kaiser, M.S. (2020). Are Bottom-Up Approaches in Development More Effective than Top-Down Approaches?. Journal of Asian Social Science Research. 2. 91-109. 10.15575/jassr.v2i1.20.

KNILL, C., DEBUS, M. and HEICHEL, S. (2010), Do parties matter in internationalised policy areas? The impact of political parties on environmental policy outputs in 18 OECD countries, 1970–2000. European Journal of Political Research, 49: 301-336. https://doi.org/10.1111/j.1475-6765.2009.01903.x

Leruth, B. (2024). Political long‐termism and the European Union: Five research questions for the future. Contemporary European Politics. 2. 10.1002/cep4.70000.

Li, Y. (2023). The Influence of Top-Down Mode and Bottom-up Mode to National Innovation and Entrepreneurship. Journal of Innovation and Development. 2. 158-161. 10.54097/jid.v2i3.7640

Li, H., Liu, S., Liew, C., Li, Y., Wayne, S., & Clark, N. (2018). An investigation on the mechanism of the increased NO_2 emissions from H_2-diesel dual fuel engine. International Journal of Hydrogen Energy, 43(7), 3837-3844. https://doi.org/10.1016/j.ijhydene.2018.01.001

Lindvall, D. (2023). What motivates urban climate leaders? A study of urban climate governance in eight Swedish municipalities. International Journal of Urban Sustainable Development, 15(1), 267-281. https://doi.org/10.1080/19463138.2023.2253755

Long, E., Carlsten, C. (2022) Controlled human exposure to diesel exhaust: results illuminate health effects of traffic-related air pollution and inform future

directions. Part Fibre Toxicol 19, 11 .
https://doi.org/10.1186/s12989-022-00450-5

MacKinnon, J. G., & Webb, M. D. (2020). Randomization inference for difference-in-differences with few treated clusters. Journal of Econometrics, 218(2), 435-450. https://doi.org/10.1016/j.jeconom.2020.04.024

Marcus, M., & Sant'Anna, P. (2021). The Role of Parallel Trends in Event Study Settings: An Application to Environmental Economics. Journal of the Association of Environmental and Resource Economists. 8. 10.1086/711509

Manski, C.F. (2011). Policy choice with partial knowledge of policy effectiveness. J Exp Criminol 7, 111-125.
https://doi.org/10.1007/s11292-010-9116-2

Mathison, S. (2005). Backward mapping. In S. Mathison (Ed.) Encyclopedia of evaluation (pp. 29-30). SAGE Publications, Inc.,
https://doi.org/10.4135/9781412950558.n42

Matthaios, V. N., Kramer, L. J., Sommariva, R., Pope, F. D., & Bloss, W. J. (2019). Investigation of vehicle cold start primary NO2 emissions inferred from ambient monitoring data in the UK and their implications for urban air quality. Atmospheric Environment, 199, 402-414.
https://doi.org/10.1016/j.atmosenv.2018.11.031

Mickwitz, P. (2003). A Framework for Evaluating Environmental Policy Instruments Context and Key Concepts. Evaluation. 9. 415-436. 10.1177/1356389003094004.

Månson, P. (2010). Båten i parken: Introduktion till samhällsstudier. Studentlitteratur.

Naturvårdsverket. (2025). Sustainable transports. https://www.naturvardsverket.se/en/international/research/the-environmental-research-fund/research-efforts-climate/sustainable-transports/

Naturvårdsverket. (1999). Samordning Och Målkonflikter: Sektorintegreringens Möjligheter Och Problem (Stockholm, Naturvårdsverket).

Nilsson, M., & Persson, Å. (2003). Framework for Analysing Environmental Policy Integration. Journal of Environmental Policy & Planning. 5. 333-359. 10.1080/1523908032000171648.

OECD. (2025). Environment at a Glance Indicators, OECD Publishing, Paris, https://doi.org/10.1787/ac4b8b89-en (accessed on %TODAY%).

Ostermeijer, F., Koster, H., van Ommeren, J., & Nielsen, V. (2022). Automobiles and urban density. Journal of Economic Geography, 22(1), 1-23. https://doi.org/10.1093/jeg/lbab047

Otola, I. (2010). Considerations of Effectiveness - Its Essence and Measurements.

Peck, J., & Tickell, A. (2002). Neoliberalizing Space. Antipode. 34. 380 - 404. 10.1111/1467-8330.00247.

Puig, D., Adger, N.W., Barnett, J., Vanhala, L., Boyd, E. (2025) Improving the effectiveness of climate change adaptation measures. Climatic Change 178, 7. https://doi.org/10.1007/s10584-024-03838-8

Purtle, J., Moucheraud, C., Yang, L. H., & Shelley, D. (2023). Four very basic ways to think about policy in implementation science. Implementation Science Communications, 4, Article 97. https://doi.org/10.1186/s43058-023-00497-1

Sabatier, P. A. (1986). Top-down and Bottom-up Approaches to Implementation Research: A Critical Analysis and Suggested Synthesis. Journal of Public Policy, 6(1), 21-48. http://www.jstor.org/stable/3998354

Sarmiento, L., Wägner, N., Zaklan, A. (2023). The air quality and well-being effects of low emission zones, Journal of Public Economics, Volume 227, 105014, ISSN 0047-2727, https://doi.org/10.1016/j.jpubeco.2023.105014

Sasabuchi, Y. (2021). Introduction to Difference-in-differences Design. Annals of Clinical Epidemiology. 3. 74-77. 10.37737/ace.3.3_74

Seelarbokus, C. (2014). Assessing the Effectiveness of International Environmental Agreements (IEAs): Demystifying the Issue of Data Unavailability. SAGE Open. 4. 10.1177/2158244014521820

Shin, H., & Lee, T. (2025). Effectiveness of social distancing under partial compliance of individuals. European Journal of Operational Research, 322(3), 949-959. https://doi.org/10.1016/j.ejor.2024.11.006

Swedish Meteorological and Hydrological Institute (SMHI). (n.d.). Yearly statistics. SMHI Data Portal. Retrieved April 3, 2026, from SMHI Data Portal Yearly Statistics

Singam, V., Zahari, H. M., & Rafiuddin, N. M. (2024). A systematic review on carbon emission of light-duty vehicles in urban environment. Social Sciences & Humanities Open, 10, Article 100924. https://doi.org/10.1016/j.ssaho.2024.100924

Singh, P., & Namrata. (2025). The role of electric vehicles and eco-friendly technologies in reducing CO2 emissions, Journal of Environmental Management, Volume 390, 126321, ISSN 0301-4797, https://doi.org/10.1016/j.jenvman.2025.126321

Skjærseth, J., & Wettestad, J. (2010). Understanding the Effectiveness of EU Environmental Policy: How Can Regime Analysis Contribute?. Environmental Politics, 11, 99-120. https://doi.org/10.1080/714000635

Smith, A., & Stirling, A. (2007). Moving Outside or Inside? Objectification and Reflexivity in the Governance of Socio-Technical Systems. Journal of Environmental Policy & Planning - J ENVIRON POL PLAN. 9. 351-373. 10.1080/15239080701622873

Smith, H., Bleibaum, R. N., & Thomas, H. A. (2012). Descriptive analysis. In H. Stone, R. N. Bleibaum, & H. A. Thomas (Eds.), Sensory evaluation practices (4th ed., pp. 233-289). Academic Press. https://doi.org/10.1016/B978-0-12-382086-0.00006-6

Stata. (2025). Stata 19 causal inference and treatment-effects estimation reference manual. Stata Press. https://www.stata.com/manuals/causalxthdidregress.pdf

Statiska Centralbyrån. (SCB). (n.d.). About SCB. https://www.scb.se/om-scb/ Data collected: 2026-04-20.

Statistiska centralbyrån (SCB). (n.d.). Statistikdatabasen: Kommunfullmäktigval - valresultat, andel röster (av giltiga röster) efter region, parti mm och valår. Retrieved May 6, 2026, from https://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START__ME__ME0104__ME0104A/ME0104T1/table/tableViewLayout1/

Stone, H., Bleibaum, R. N., & Thomas, H. A. (2012). Descriptive analysis. In H. Stone, R. N. Bleibaum, & H. A. Thomas (Eds.), Sensory evaluation practices (4th ed., pp. 233-289). Academic Press. https://doi.org/10.1016/B978-0-12-382086-0.00006-6

Stoker, G. (1998). Theory and Urban Politics. International Political Science Review / Revue Internationale de Science Politique, 19(2), 119-129. http://www.jstor.org/stable/1601319

Sutherland, C. (2025). Urban reform coalitions for transformative sustainability. https://doi.org/10.1177/09562478251332260

Taha, S. S., Idoudi, S., Alhamdan, N., Ibrahim, R. H., Surkatti, R., Amhamed, A., & Alrebei, O. F. (2025). Comprehensive review of health impacts of the exposure to nitrogen oxides (NOx), carbon dioxide (CO2), and particulate

matter (PM). Journal of Hazardous Materials Advances, 19, 100771. https://doi.org/10.1016/j.hazadv.2025.100771

Tafidis, P., Gholamnia, M., Sajadi, P., Krishnan Vijayakrishnan, S., & Pilla, F. (2024). Evaluating the impact of urban traffic patterns on air pollution emissions in Dublin: A regression model using Google Project Air View data and traffic data. European Transport Research Review, 16(1), Article 47. https://doi.org/10.1186/s12544-024-00671-z

Thana Singam, V., Zahari, H. M., & Rafiuddin, N. M. (2024). A systematic review on carbon emission of light-duty vehicles in urban environment. Social Sciences & Humanities Open, 10, Article 100924. https://doi.org/10.1016/j.ssaho.2024.100924

Transportstyrelsen. (2024). Miljözoner. https://www.transportstyrelsen.se/sv/vagtrafik/miljo/miljozoner/ (accessed 19 March 2026).

Transportstyrelsen. (2024). Miljözoner. November 22. https://www.transportstyrelsen.se/sv/vagtrafik/miljo/miljozoner/

Tögel, M., & Špička, L. (2014). Low-Emission Zones in European Countries. Transactions on Transport Sciences. 7. 97-108. 10.2478/trans-2014-0007.

Umeå kommun. (2020). Miljözoner i Sverige [PDF]. Umeå kommun. https://www.umea.se/download/18.250f9659174ae4b979462d/1600932019086/Milj%C3%B6zoner%20i%20Sverige.pdf

United Nations. (UN). (2023). Tackling urban air quality (online). Available at: https://unece.org/tackling-urban-air-quality (accessed 19 March 2026).

Valmyndigheten. (2025). Riksdags-, region- och kommunval - resultat. https://www.val.se/valresultat-och-statistik/riksdags--region--och-kommunval/tidigare-valresultat

Van der Heijden, J. (2019(. Studying urban climate governance: Where to begin, what to look for, and how to make a meaningful contribution to scholarship and practice. Earth Sys Govern. 1(May):100005. doi: 10.1016/j.esg.2019.100005

Vetenskapsrådet. (2002). Forskningsetiska principer inom humanistisk-samhällsvetenskaplig forskning. https://www.vr.se/uppdrag/etik.html. (Accessed 2026-05-29).

Vince, J., Fudge, M., Fullbrook, L., & Haward, M. (2024). Understanding policy integration through an integrative capacity framework. Policy and Society, 43(3), 381–395. https://doi.org/10.1093/polsoc/puae027

Warner, M. E. (2012). Privatization and urban governance: The continuing challenges of efficiency, voice, and integration. Cities, 29(Suppl. 2), S38-S43. https://doi.org/10.1016/j.cities.2012.06.007

Wright, B., & Wallis, S. E. (2019). Backward mapping. In B. Wright & S. E. Wallis (Eds.), Practical mapping for applied research and program evaluation. SAGE Publications. https://sk.sagepub.com/ency/edvol/evaluation/chpt/backward-mapping

Wu, L., Yuan, X., Yin, C., Yang, M., & Ouyang, H. (2023). Car ownership behavior model considering nonlinear impacts of multi-scale built environment characteristics. Sustainability, 15, 9658. https://doi.org/10.3390/su15129658

Zhang, K., & Batterman, S. (2013). Air pollution and health risks due to vehicle traffic, Science of The Total Environment, Volumes 450-451, Pages 307-316, ISSN 0048-9697, https://doi.org/10.1016/j.scitotenv.2013.01.074

9. Appendix

Table 3. Event study, effects of Low emission zones on privately owned cars per 1000 inhabitants without control variables.

Event time	**Estimate**	**Std. Error**	**95% Simultaneous Confidence Band**
-7	-0.6667	0.6589	[-2.0000, 0.6667]
-6	-1.8333	1.7297	[-5.3333, 1.6667]
-5	-0.5000	1.1120	[-2.7500, 1.7500]
-4	-1.3333	1.3179	[-4.0000, 1.3333]
-3	-0.1667	1.5650	[-3.3333, 3.0000]
-2	-2.5000	1.2355	[-5.0000, 0.0000]
-1	-0.7778	1.4826	[-3.7778, 2.2222]
0	-3.4444*	0.5491	[-4.5556, -2.3333]
1	-4.7778*	0.7138	[-6.2222, -3.3333]
2	-5.5556*	0.5491	[-6.6667, -4.4444]
3	-6.8889*	0.5491	[-8.0000, -5.7778]
4	-9.8889*	1.8670	[-13.6667, -6.1111]
5	-12.5556*	3.0750	[-18.7778, -6.3333]
6	-15.6667*	4.6125	[-25.0000, -6.3333]
7	-19.5000*	2.5946	[-24.7500, -14.2500]
8	-17.6667*	1.1531	[-20.0000, -15.3333]
9	-19.3333*	0.8237	[-21.0000, -17.6667]

10	-17.3333*	0.3295	[-18.0000, -16.6667]
11	-19.0000*	3.9536	[-27.0000, -11.0000]
12	-18.6667*	3.6241	[-26.0000, -11.3333]

*p<0,05, Standard errors are clustered on Regional level

Table 4. Event study, effects of Low emission zones on average Nitrogen dioxide emissions without control variables.

Event time	**Estimate**	**Std. Error**	**95% Pointwise Confidence Band**
-7	4.5775*	1.8199	[1.0106, 8.1444]
-6	-4.2750	NA	NA
-5	0.4650	5.8933	[-11.0857, 12.0157]
-4	0.1650	1.7013	[-3.1695, 3.4995]
-3	-2.3825	3.8566	[-9.9413, 5.1763]
-2	-7.3925*	0.6913	[-8.7474, -6.0376]
-1	-2.5075	2.3610	[-7.1351, 2.1201]
0	5.2600*	2.1498	[1.0465, 9.4735]
1	-1.9725*	0.5597	[-3.0695, -0.8755]
2	-4.5550	2.8762	[-10.1923, 1.0823]
3	0.5338	2.1609	[-3.7015, 4.7690]
4	-0.7000	3.7139	[-7.9791, 6.5791]
5	-4.5062*	1.5373	[-7.5193, -1.4932]
6	-6.1062*	1.3140	[-8.6816, -3.5309]

7	-2.4675*	1.0063	[-4.4398, -0.4952]
8	-3.3275*	0.4503	[-4.2102, -2.4448]
9	-5.7075*	0.9396	[-7.5491, -3.8659]
10	-6.8525*	1.5363	[-9.8637, -3.8413]
11	-4.5525*	0.2057	[-4.9557, -4.1493]
12	-6.1975*	0.4707	[-7.1201, -5.2749]

*p<0,05, Standard errors are clustered on Regional level

Table 5. Event study, effects of Low emission zones on average number of heavy lorries without control variables.

Event time	**Estimate**	**Std. Error**	**95% Simultaneous Confidence Band**
-7	0.0000	18.2854	[-343.4359, 343.4359]
-6	2.8333	34.9235	[-653.0983, 658.7650]
-5	-10.0000	41.0187	[-780.4103, 760.4103]
-4	-49.5000	37.0651	[-745.6538, 646.6538]
-3	-67.8333	38.4241	[-789.5128, 653.8462]
-2	14.6667	15.6497	[-279.2650, 308.5983]

-1	41.8889	9.0054	[-127.2507, 211.0285]
0	19.6667	24.3806	[-438.2479, 477.5812]
1	61.0000	59.3041	[-1052.8462, 1174.8462]
2	21.5556	29.2128	[-527.1168, 570.2279]
3	12.3333	66.7171	[-1240.7436, 1265.4103]
4	64.7778	52.6049	[-923.2450, 1052.8006]
5	60.6667	71.8238	[-1288.3248, 1409.6581]
6	95.2222	71.0551	[-1239.3305, 1429.7749]
7	159.5000	91.7978	[-1564.6410, 1883.6410]
8	104.3333	31.7936	[-492.8120, 701.4786]
9	228.3333	52.0558	[-749.3761, 1206.0427]
10	330.0000	43.4897	[-486.8205, 1146.8205]
11	443.0000	75.1185	[-967.8718, 1853.8718]

12	283.3333	56.0094	[-768.6325, 1335.2991]

*p<0,05, Standard errors are clustered on Regional level

Table 6. The regions and their (code) that are used and analysed in the study

Sweden (00), Stockholm county (01), Uppsala county (03), Södermanland county (04), Östergötland county (05), Jönköping county (06), Kronoberg county (07), Kalmar county (08), Gotland county (09), Blekinge county (10), Skåne county (12), Halland county (13), Västra Götaland county (14), former Älvsborg county (15), former Skaraborg county (16), Värmland county (17), Örebro county (18), Västmanland county (19), Dalarna county (20), Gävleborg county (21), Västernorrland county (22), Jämtland county (23), Västerbotten county (24), Norrbotten county (25), Upplands Väsby (0114), Vallentuna (0115), Österåker (0117), Värmdö (0120), Järfälla (0123), Ekerö (0125), Huddinge (0126), Botkyrka (0127), Salem (0128), Haninge (0136), Tyresö (0138), Upplands-Bro (0139), Nykvarn (0140), Täby (0160), Danderyd (0162), Sollentuna (0163), Stockholm (0180), Södertälje (0181), Nacka (0182), Sundbyberg (0183), Solna (0184), Lidingö (0186), Vaxholm (0187), Norrtälje (0188), Sigtuna (0191), Nynäshamn (0192), Håbo (0305), Älvkarleby (0319), Knivsta (0330), Heby (0331), Tierp (0360), Uppsala (0380), Enköping (0381), Östhammar (0382), Vingåker (0428), Gnesta (0461), Nyköping (0480), Oxelösund (0481), Flen (0482), Katrineholm (0483), Eskilstuna (0484), Strängnäs (0486), Trosa (0488), Ödeshög (0509), Ydre (0512), Kinda (0513), Boxholm (0560), Åtvidaberg (0561), Finspång (0562), Valdemarsvik (0563), Linköping (0580), Norrköping (0581), Söderköping (0582), Motala (0583), Vadstena (0584), Mjölby (0586), Aneby (0604), Gnosjö (0617), Mullsjö (0642),

Habo (0643), Gislaved (0662), Vaggeryd (0665), Jönköping (0680), Nässjö (0682), Värnamo (0683), Sävsjö (0684), Vetlanda (0685), Eksjö (0686), Tranås (0687), Uppvidinge (0760), Lessebo (0761), Tingsryd (0763), Alvesta (0764), Älmhult (0765), Markaryd (0767), Växjö (0780), Ljungby (0781), Högsby (0821), Torsås (0834), Mörbylånga (0840), Hultsfred (0860), Mönsterås (0861), Emmaboda (0862), Kalmar (0880), Nybro (0881), Oskarshamn (0882), Västervik (0883), Vimmerby (0884), Borgholm (0885), Gotland (0980), Olofström (1060), Karlskrona (1080), Ronneby (1081), Karlshamn (1082), Sölvesborg (1083), Svalöv (1214), Staffanstorp (1230), Burlöv (1231), Vellinge (1233), Östra Göinge (1256), Örkelljunga (1257), Bjuv (1260), Kävlinge (1261), Lomma (1262), Svedala (1263), Skurup (1264), Sjöbo (1265), Hörby (1266), Höör (1267), Tomelilla (1270), Bromölla (1272), Osby (1273), Perstorp (1275), Klippan (1276), Åstorp (1277), Båstad (1278), Malmö (1280), Lund (1281), Landskrona (1282), Helsingborg (1283), Höganäs (1284), Eslöv (1285), Ystad (1286), Trelleborg (1287), Kristianstad (1290), Simrishamn (1291), Ängelholm (1292), Hässleholm (1293), Hylte (1315), Halmstad (1380), Laholm (1381), Falkenberg (1382), Varberg (1383), Kungsbacka (1384), Härryda (1401), Partille (1402), Öckerö (1407), Stenungsund (1415), Tjörn (1419), Orust (1421), Sotenäs (1427), Munkedal (1430), Tanum (1435), Dals-Ed (1438), Färgelanda (1439), Ale (1440), Lerum (1441), Vårgårda (1442), Bollebygd (1443), Grästorp (1444), Essunga (1445), Karlsborg (1446), Gullspång (1447), Tranemo (1452), Bengtsfors (1460), Mellerud (1461), Lilla Edet (1462), Mark (1463), Svenljunga (1465), Herrljunga (1466), Vara (1470), Götene (1471), Tibro (1472), Töreboda (1473), Göteborg (1480), Mölndal (1481), Kungälv (1482), Lysekil (1484), Uddevalla (1485), Strömstad (1486), Vänersborg (1487), Trollhättan (1488), Alingsås (1489), Borås (1490), Ulricehamn (1491), Åmål (1492), Mariestad (1493), Lidköping (1494), Skara (1495), Skövde (1496), Hjo (1497), Tidaholm (1498), Falköping (1499), Kil (1715), Eda (1730), Torsby (1737), Storfors (1760), Hammarö (1761),

Munkfors (1762), Forshaga (1763), Grums (1764), Årjäng (1765), Sunne (1766), Karlstad (1780), Kristinehamn (1781), Filipstad (1782), Hagfors (1783), Arvika (1784), Säffle (1785), Lekeberg (1814), Laxå (1860), Hallsberg (1861), Degerfors (1862), Hällefors (1863), Ljusnarsberg (1864), Örebro (1880), Kumla (1881), Askersund (1882), Karlskoga (1883), Nora (1884), Lindesberg (1885), Skinnskatteberg (1904), Surahammar (1907), Heby (1917), Kungsör (1960), Hallstahammar (1961), Norberg (1962), Västerås (1980), Sala (1981), Fagersta (1982), Köping (1983), Arboga (1984), Vansbro (2021), Malung-Sälen (2023), Gagnef (2026), Leksand (2029), Rättvik (2031), Orsa (2034), Älvdalen (2039), Smedjebacken (2061), Mora (2062), Falun (2080), Borlänge (2081), Säter (2082), Hedemora (2083), Avesta (2084), Ludvika (2085), Ockelbo (2101), Hofors (2104), Ovanåker (2121), Nordanstig (2132), Ljusdal (2161), Gävle (2180), Sandviken (2181), Söderhamn (2182), Bollnäs (2183), Hudiksvall (2184), Ånge (2260), Timrå (2262), Härnösand (2280), Sundsvall (2281), Kramfors (2282), Sollefteå (2283), Örnsköldsvik (2284), Ragunda (2303), Bräcke (2305), Krokom (2309), Strömsund (2313), Åre (2321), Berg (2326), Härjedalen (2361), Östersund (2380), Nordmaling (2401), Bjurholm (2403), Vindeln (2404), Robertsfors (2409), Norsjö (2417), Malå (2418), Storuman (2421), Sorsele (2422), Dorotea (2425), Vännäs (2460), Vilhelmina (2462), Åsele (2463), Umeå (2480), Lycksele (2481), Skellefteå (2482), Arvidsjaur (2505), Arjeplog (2506), Jokkmokk (2510), Överkalix (2513), Kalix (2514), Övertorneå (2518), Pajala (2521), Gällivare (2523), Älvsbyn (2560), Luleå (2580), Piteå (2581), Boden (2582), Haparanda (2583), Kiruna (2584).

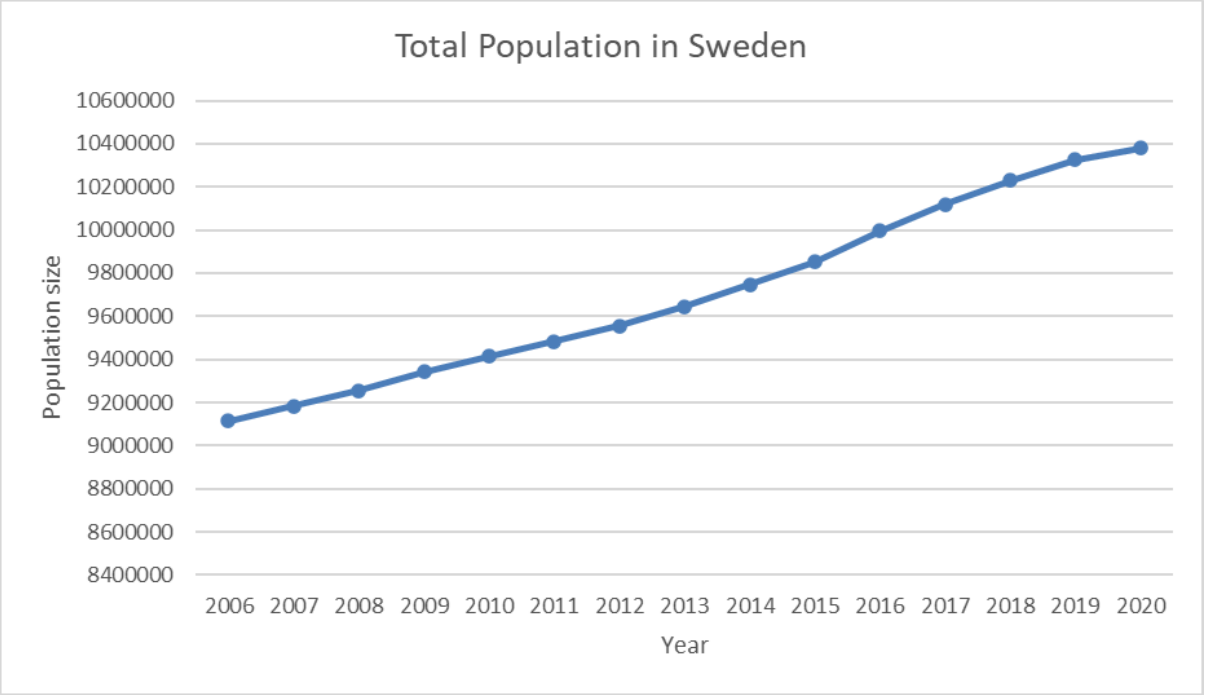


Figure 16. Population size in December of each year, from 2006-2020.